

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies
FIIT-16768-127206

Martin Kubiš
Cross-Lingual Citation Verification Using
Multilingual Transformer Models
Bachelor Thesis

Supervisor: PaedDr. Pavol Batačík

January 2026

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Study programme: Informatics

Study field: Computer Science

Place of elaboration: Institute of Computer Engineering and Applied Informatics

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ZADANIE BAKALÁRSKEJ PRÁCE

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Názov práce: **Nástroj na rozpoznávanie textov z cudzojazyčných prekladov**

Jazyk, v ktorom sa práca
vypracuje: slovenský jazyk

Špecifikácia zadania: V prostredí rastúcej dostupnosti zahraničných vedeckých výstupov a rozvoja prekladových technológií je čoraz väčšie riziko zneužitia cudzích odborných textov formou tzv. prekladového plagiátorstva. Tento jav spočíva v preklade vedeckých článkov alebo iných odborných prác z cudzieho jazyka do slovenčiny (alebo iného cieľového jazyka) bez riadneho uvedenia pôvodného autora a zdroja, pričom sa text následne vydáva za vlastný výskum. Takéto konanie je potrebné považovať za plagiát v plnom rozsahu, keďže dochádza k porušeniu autorských práv a etických zásad vedeckej činnosti. V tomto kontexte analyzujte všetky dostupné existujúce riešenia na identifikáciu sémantickej zhody medzi textami. Navrhnite a implementujte vlastnú interpretáciu realizácie rozpoznania sémantickej zhody aspoň pre tri jazyky. Tento rámec pre detekciu preložených plagiátov – napr. porovnávanie významovej podobnosti (semantic similarity), využitie embeddingov (napr. Word2Vec, BERT, Sentence-BERT) alebo spätného prekladu, ktorý umožní porovnanie originálneho textu so suspektným textom v inom jazyku evalvujte nad datasetom netriviálneho obsahu.

Rozsah práce: 40

Termín odovzdania práce: 11. 05. 2026

Declaration of Honour

I declare on my honour that I wrote this thesis independently under the supervision of PaedDr. Pavol Batalík and with the use of the cited literature.

In Bratislava, 1.1.2026

Martin Kubiš

Acknowledgment

I would like to thank my thesis supervisor, PaedDr. Pavol Batalík, for his guidance, feedback, and support during the preparation of this bachelor's thesis.

Annotation

Slovak University of Technology Bratislava

FACULTY OF INFORMATICS AND INFORMATION TECHNOLOGIES

Study program: Informatics

Author: Martin Kubiš

Bachelor's thesis: Cross-Lingual Citation Verification Using Multilingual
Transformer Models

Supervisor: PaedDr. Pavol Batalík

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This bachelor's thesis focuses on cross-lingual citation verification in academic writing. The problem arises when a cited source is written in another language and the citing author reuses the source through translation or paraphrase in a way that may no longer faithfully reflect the original meaning. Conventional plagiarism detection systems are effective mainly for monolingual overlap and are therefore poorly suited to this task. The thesis reviews the legal and technical context of plagiarism and citation misuse, summarizes related work in cross-lingual plagiarism detection and citation verification, and discusses multilingual transformer models suitable for English-Slovak semantic comparison. The current contribution is a structured proposal for a multi-stage verification pipeline that extracts citations and references from PDF documents, retrieves the cited sources, segments the source text, and compares cited statements with source passages using multilingual embeddings. The thesis also identifies the main risks and limitations of the approach.

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1 Technical Abstract

This bachelor's thesis addresses the problem of cross-lingual citation verification in academic writing. The problem arises when an author cites a source written in another language and then reuses its content through translation or paraphrase. In such cases, standard plagiarism detection systems often fall short because they focus mainly on surface overlap and are not designed to assess whether the cited claim is semantically supported by the original source. This leaves a gap between plagiarism detection and citation verification.

The thesis outlines the legal and ethical context of plagiarism, reviews prior work on cross-lingual plagiarism detection and citation verification, and compares multilingual transformer-based approaches suitable for semantic comparison across languages.

The main result is a proposal for a multi-stage verification pipeline. The proposed system takes a PDF document as input, extracts in-text citations and bibliography entries, retrieves the cited source documents, segments the retrieved text, and compares cited claims with source passages using multilingual embeddings. The intended output is a protocol-style report that supports human assessment by highlighting the passages most relevant to each citation.

2 Lay Summary

This thesis is about checking whether academic citations are actually used honestly when the original source is written in a different language. A student may cite an English paper in a Slovak text, translate the idea, and slightly rewrite it. Even if a citation is present, the rewritten sentence may not match what the source really says. Ordinary plagiarism tools usually compare words and phrases, so they often miss this kind of problem.

The goal of the thesis is to explore how a computer system could help with this task. The proposed tool would read a thesis in PDF form, detect the citations and the bibliography, try to find the original sources, and then compare the cited statements with the source text. To do that, it would use multilingual language models that can compare meaning across languages rather than just identical wording.

3 Introduction

Plagiarism is commonly defined as presenting another person’s writing as one’s own, a form of misconduct closely connected with forgery and piracy[1]. In academic practice, however, the issue goes beyond simple copy-paste. It also includes uncredited reuse of ideas, misleading paraphrase, false attribution, and other forms of source misuse discussed in the plagiarism-detection literature[2]. Anderson and Steneck[3] emphasize that the ethical problem begins when borrowed ideas are reused without proper acknowledgment, not when researchers merely learn from prior work. That distinction matters because it shows why plagiarism is not only a legal issue but also a question of scholarly honesty.

In the Slovak academic environment, this problem is grounded both in law and in university practice. The Higher Education Act[4] requires that a submitted thesis be, at least in part, the result of the student’s own independent work. The Copyright Act[5] protects original works as results of creative intellectual activity and grants the author both moral and economic rights, including the right to be identified as the creator and to control the use of the work. These legal rules address obvious forms of misuse, but they do not by themselves answer the technical question of how such misuse can be detected in modern digital documents.

The difficulty becomes even greater when the source and the suspicious document are written in different languages. Cross-lingual plagiarism detection studies cases in which a document in one language reuses material from a source in another language, either through direct translation or paraphrased translation[6]. Citation verification is related but more specific: it asks whether a cited source actually supports the claim made in the citing text[7]. A sentence may contain a formally correct citation and still misrepresent the source after translation, simplification, or paraphrase. That makes cross-lingual citation misuse harder to detect than ordinary monolingual copying.

Recent work suggests that multilingual semantic models can help narrow this gap. Research in cross-lingual plagiarism detection has moved from string matching and translation resources toward embedding-based approaches built on transformer architectures such as mBERT[8] and XLM-R[9]. At the same time, citation-verification research is starting to explore claim-evidence modelling[10], full-text evidence retrieval[11], and large language models for assessing whether a cited claim is supported by a source passage[7].

4 Problem Definition

In a typical scenario, an author cites an English article in a Slovak thesis, translates the relevant claim, paraphrases it, and presents the result with a formally correct reference. The citation may look proper on the surface, yet the translated statement may oversimplify, distort, or even contradict the original meaning. This kind of misuse is harder to detect than literal copying because the wording changes while the argumentative role of the source remains central[6].

This problem belongs to the broader domain of plagiarism, which is not limited to direct copy-paste behaviour. A widely cited taxonomy distinguishes deliberate copy-paste plagiarism, simple paraphrasing, patchwork paraphrasing, self-plagiarism, illegitimate-source plagiarism, retweet plagiarism, metaphor plagiarism, and idea plagiarism[12]. In computer science, the same paper also highlights source-code plagiarism categories such as local manipulation, structure reordering, whitespace-only disguise, and even language switching, all of which show that superficial modifications are often used to evade detection[12].

Cross-lingual plagiarism itself is usually defined as reuse in which a suspicious document in one language originates from a document in another language, either through exact translation or paraphrased translation[13]. Prior work From Ehsan et. al[13] distinguishes syntax-based, dictionary-based, parallel-corpus-based, comparable-corpus-based, and multilingual semantic-network-based strategies for comparing such texts.

The practical importance of the problem is considerable. Foltýnek et al.[14] note that “even in the best case, i.e., if the plagiarism is discovered, reviewing and punishing plagiarized research papers and grantapplications still causes a high effort for the reviewers, affected institutions, and funding agencies”. If it remains undiscovered, plagiarists may gain academic credit, funding, or career advancement based on misappropriated work[14]. In the case of citation verification, the risk is not only theft of text but also the spread of unsupported or misleading claims in academic writing. A system that can automatically assist with cross-lingual citation checking would therefore address both integrity and quality-control concerns.

The goal of this thesis is not to solve general plagiarism detection. Instead, it aims to define a focused verification pipeline for academic texts, in which the system identifies citations, retrieves sources, and estimates whether the cited source semantically supports the corresponding claim.

5 Overview of Related Solutions

The most mature neighbouring area is cross-lingual plagiarism detection. Early work relied on dictionary-based translations[15] and syntax-based approaches[16], but these tend to be computationally expensive and limited by available resources. Classical retrieval-oriented or lexical methods, including string matching[17] and n-gram[18] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[19]. Amirzhanov et al.[19] note that this limitation has pushed the field away from symbolic and lexical strategies toward semantic representations better suited to multilingual comparison.

The introduction of multilingual neural models changed the research landscape. The first major multilingual transformer to gain widespread use was mBERT[8]. BERT itself introduced masked language modelling, in which some words are hidden and the model is trained to predict them from context. By training on very large corpora, the model learns rich contextual embeddings that capture meaning rather than simple word identity. XLM-R[9] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[20]-style training improvements such as dynamic masking and larger batches. Newer families such as XLM-V[21], ModernBERT[22], and mmBERT[23] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency. Recent studies report strong results for transformer-based multilingual embeddings in semantic comparison and paraphrase-oriented cross-lingual settings[24].

Commercial systems remain strong in monolingual academic workflows but weak in cross-lingual use cases. Turnitin[25] and iThenticate[26] are widely deployed because they can match suspicious passages against very large indexed repositories and generate practical reports for institutions. Regional platforms such as Odevzdej[27] and CRZP[28] play a similar role in Czech and Slovak environments. Their main limitation is that they are designed primarily for detecting direct or near-direct overlap within a single language. They are not built for multilingual environments and they function as plagiarism detectors rather than citation verification systems.

Citation verification is less mature than plagiarism detection, but several works are directly relevant to this thesis. Some research focuses on citation intent and rhetorical function, as in SciCite[29]. A much closer conceptual match is the study by Zeeb and Olsson[7], which explicitly asks whether large language models can evaluate citations in scholarly documents. They treat citation evaluation as a

binary classification problem, using ChatGPT to assign a boolean result together with a short explanation[7]. Although the approach by Zeeb and Olsson was not designed as a practical system, it shows that LLMs can perform the core reasoning task of citation verification.

An even more direct systems-level point of comparison is Haan’s[11] SemanticCite, which proposes citation verification through AI-powered full-text analysis and evidence-based reasoning. SemanticCite is especially relevant because it moves beyond abstract discussion of citation quality and treats the task as a pipeline problem. The system must work with full-text documents, identify citations, retrieve and analyse source material, and produce evidence that can justify the verification result[11].

The need for citation-verification tools is not new. Wright and Armstrong[30] described citation accuracy as a structural weakness in scholarly communication. Aronsky et al.[31] quantified substantial reference errors even in biomedical informatics journals, where almost one-third of citations contained mistakes. This underlines the value of automated tools for checking citations. Large language models are now also being tested in neighbouring tasks such as scientific-misinformation detection[10], but these approaches still raise questions of reproducibility, cost, and consistency.

Recent work also warns that threshold choices and dataset bias can distort reported performance, especially in imbalanced settings[32][33]. These observations matter here because the evaluation of any proposed system will face similar trade-offs.

6 Solution Overview

The proposed system is meant to support citation checking, not to replace human judgment. In practice, verifying a citation is often routine work: find the cited source, locate the relevant passage, and decide whether it really supports the claim made in the thesis. The goal of the solution is to automate these repetitive steps and return enough evidence for the evaluator to make the final decision.

For that reason, the problem is treated as a pipeline rather than as a single classifier. The system has to work first with a PDF document, then with bibliography metadata, and finally with short claim-evidence pairs. These are different kinds of data, so it makes more sense to handle them in separate stages. A pipeline also makes failures easier to interpret: if a citation cannot be matched, a source can-

not be retrieved, or the evidence is weak, that information should remain visible in the output instead of being hidden behind a single score. In this sense, the design is close to SemanticCite, which also approaches citation verification as an evidence-based process[11].

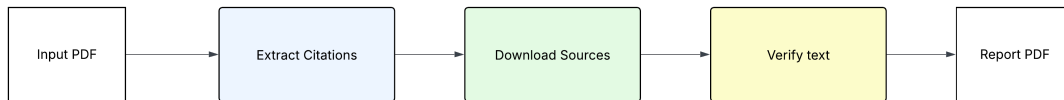


Figure 1: High-level pipeline of the proposed system

Figure 1 shows the overall logic. The input is an academic PDF document. The system extracts the citations actually used in the text, retrieves the referenced sources, compares the citing sentences with candidate passages from those sources, and produces a protocol-style PDF report. The output is therefore not just a verdict, but a trace of how the citation was checked.

6.0.1 Architecture and design choices

The architecture is divided into three stages: document understanding, source acquisition, and semantic verification. Figure 2 makes this structure explicit. It also reflects the actual workflow of a reviewer: first identify what is being cited, then obtain the cited material, and only then ask whether the source supports the claim.

The detailed pipeline also shows an important design decision: the system does not compare whole documents only. Citation misuse usually appears locally, at the level of a sentence or a short passage, so the comparison has to stay tied to individual citation contexts rather than to document-level similarity.

6.0.2 Document understanding and reference alignment

The first stage processes the submitted PDF and extracts two layers in parallel: in-text citations and bibliography entries. These two outputs then have to be aligned, because later verification depends on linking each citation in the body text to a concrete item in the reference list. This citation-reference pair becomes the basic unit for the rest of the analysis.

The system is deliberately limited to IEEE-style numeric citations. Square-bracket references are easier to detect reliably than author-year formats, footnotes, or

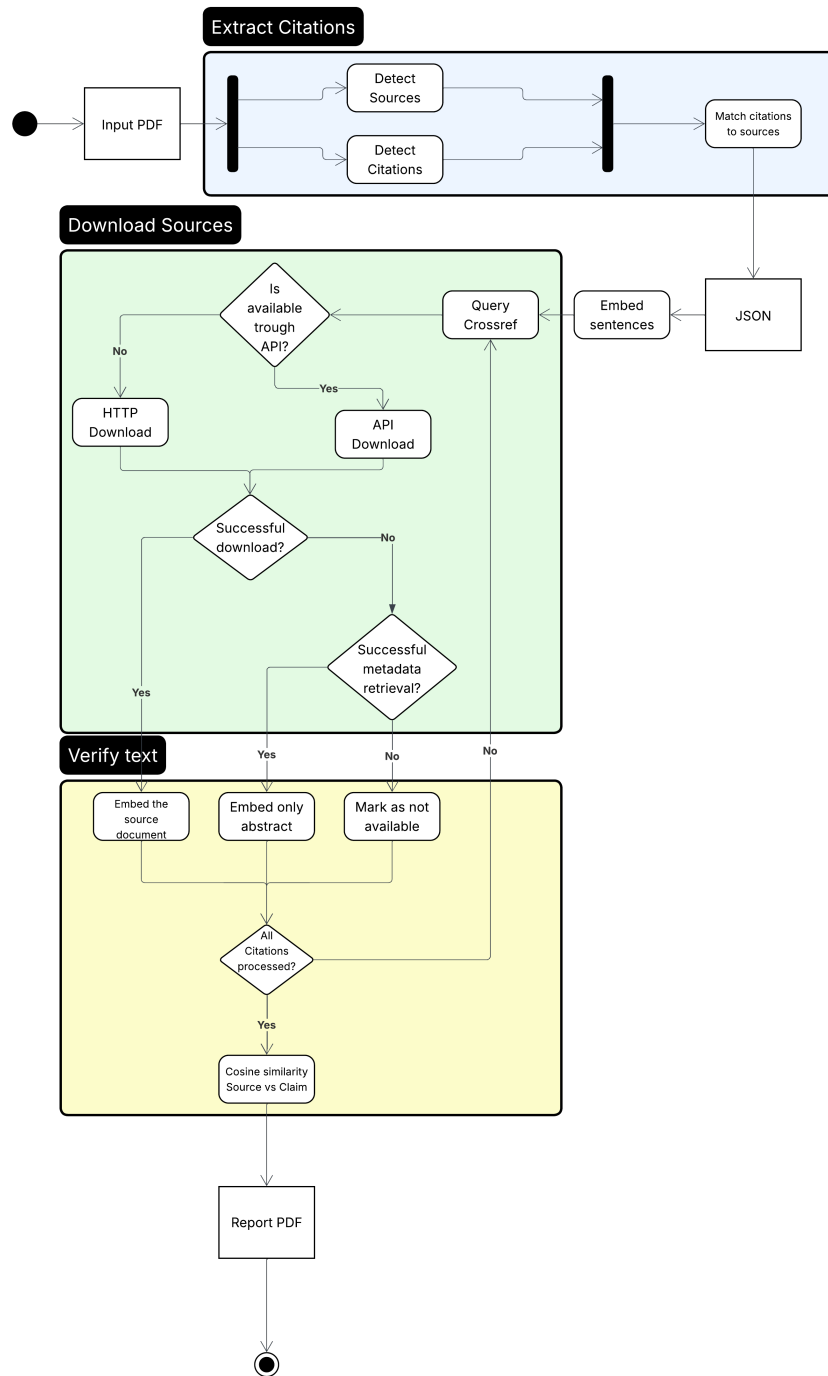


Figure 2: Detailed processing pipeline of the proposed system

mixed institutional conventions, so this restriction keeps the focus on citation verification itself rather than on turning citation-style parsing into a separate problem.

After detection, each reference is converted into a structured internal representation containing the original reference string, available identifiers such as DOI, the citing sentences, and the number of occurrences in the text.

This stage also reveals one of the main limitations of the whole solution. PDF files are not designed for reliable machine interpretation. Bibliography sections may appear under different headings, line breaks can disturb sentence boundaries, and OCR noise can damage both citation detection and reference parsing. If the document is parsed incorrectly at this point, the later stages inherit that error.

6.0.3 Source retrieval and evidence preparation

Once the system identifies the references actually used in the text, it attempts to obtain the corresponding source documents. Retrieval follows a layered strategy. If a DOI is present in the reference, the system uses it in a Crossref API[34] request to obtain precise document metadata. If no DOI is available, it falls back to Crossref's full-text search. If neither approach returns a reliable match, the source is marked as unavailable.

When Crossref returns a probable match, the system then attempts to retrieve the document itself. It first checks the DOI and related metadata to determine whether the full text is accessible through one of the publisher APIs available to the system. If that is not possible, it falls back to standard HTTP requests. When the full text cannot be obtained, the system still tries to retrieve at least the abstract. If no usable source can be found, the reference remains marked as unavailable.

This is preferable to a single retrieval mechanism because academic sources vary widely in accessibility. Some are open, some expose only metadata, and others are blocked by publisher restrictions. Retrieval failure is therefore informative in its own right and should be reported explicitly rather than silently ignored.

After a source has been obtained, the text is segmented into smaller units (sentences, paragraphs, or sections) using GROBID[35]. This step is necessary because citation support tends to be localised. A paper may address the right topic in general while only specific passages directly support the claim attributed to

it. Passage-level comparison is therefore more informative than whole-document similarity[36].

6.0.4 Semantic verification

The central analytical stage maps both the citing sentences and the retrieved source passages into a shared multilingual semantic space. Similarity is then computed between a claim in the thesis and candidate evidence passages using cosine similarity. This is suitable for cross-lingual verification because it does not rely on exact wording overlap and does not assume literal translation[24].

In the implemented design, this comparison is based on multilingual transformer embeddings, with XLM-R as the main model because it offers broad language coverage and solid performance in cross-lingual semantic tasks. The purpose of this stage is not to declare a citation true or false with certainty, but to rank the most relevant passages and show where support appears strongest.

6.0.5 Reporting, limitations, and interdisciplinary impact

The output of the pipeline is a protocol-style PDF report. It summarises the detected references, their frequency of use, unmatched citations, unavailable sources, and the passages that appear most relevant to each checked claim. This is more useful than a single numerical score because citation problems are heterogeneous and require context.

The main limitations remain straightforward. The system supports only one citation style, it depends heavily on PDF extraction quality and source availability, and its semantic component may still be weaker for Slovak or for highly specialised academic terminology than for high-resource English material. Similarity thresholds also have to be calibrated carefully, since poor calibration can make the output look more certain than it really is[33].

The system should not be understood as a fully automatic truth-assessment tool. High similarity suggests support, but it does not by itself prove correct interpretation. Low similarity is also not enough to prove misuse, because a valid citation may summarise a broader argument or depend on context outside a single sentence. The role of the system is therefore to narrow the search space for the human assessor, not to replace that assessor.

7 Risk Assessment

This project carries both technical and practical risks, but most of them can be reduced by keeping the scope narrow and by treating the system as decision support rather than as full automation.

Strengths. The main strength of the project is its focused scope. Instead of attempting to solve general multilingual plagiarism detection, the work concentrates on citation verification, which is easier to define, implement, and evaluate. A second strength is the modular pipeline design. Citation extraction, source retrieval, and semantic verification are separated, which makes errors easier to trace and the system easier to extend.

Weaknesses. The main weakness is the system’s strong dependence on input quality. PDF extraction can fail because of inconsistent formatting, OCR noise, or unusual bibliography structure. The solution is also limited to IEEE-style numeric citations, which improves reliability but reduces generality. Another weakness is its dependence on pretrained multilingual models, whose performance may be lower for low-resource languages or for highly specialised academic language[37]. Domain mismatch creates an additional problem because pretrained language models are often trained on general web data rather than academic prose[38]. Finally, the system does not produce a final verdict, only evidence, which may be less satisfying for users who want a clear true/false answer.

Opportunities. The project addresses a practical problem that is not well covered by standard plagiarism detection tools. If the core pipeline proves useful, it could be extended with a better interface, broader citation-style support, or deployment as an online service for schools. The same architecture could also be reused for other tasks in which claims need to be linked with source passages across documents.

Threats and mitigation. The most important external threat is source availability. Many cited works cannot be downloaded as full text because of publisher restrictions or broken links. This risk cannot be removed completely, so the mitigation strategy is to use layered retrieval, fall back to abstracts when possible, and report unavailable sources explicitly. Another threat is semantic uncertainty in cross-lingual comparison. Similarity scores may be affected by translation differences, paraphrasing, domain-specific terminology, or idiomatic expressions[39]. To reduce this risk, the system does not produce a final verdict and instead presents evidence for human review. A final risk is scope growth. The topic could easily expand into citation parsing, web application development, or full plagiarism detection. This

was mitigated by restricting the thesis to one citation style and one clearly defined verification workflow.

8 Experimental Reproducibility and Impact

8.0.1 Dataset and experimental scope

The main experiment processed a corpus of 1000 bachelor and master theses, with 250 documents each from four faculties focused on Computer Science or Software Engineering across Slovak universities. All source documents were downloaded from the Central Register of Theses and Dissertations (*CRZP*[28]), a publicly accessible portal that archives all theses submitted at accredited Slovak higher education institutions.

Although the individual PDF files are publicly reachable through CRZP, the dataset assembled for this study is not redistributed. The students whose work was analysed did not give explicit consent for their theses to be used as research data, so sharing the collected corpus would conflict with reasonable privacy expectations even where no formal restriction exists.

The four faculties sampled are: Faculty of Informatics and Information Technologies STU Bratislava, Faculty of Electrical Engineering and Information Technology STU Bratislava, Faculty of Electrical Engineering and Information Technology UNIZA Žilina, and the Informatics program at Faculty of Mathematics, Physics and Informatics UK Bratislava. All theses were submitted between 2020 and 2025.

In the comparative figures, these are labelled Faculty 1 through Faculty 4 to avoid direct attribution of results to a specific institution.

Researchers who wish to replicate the study can reconstruct an equivalent dataset by querying CRZP directly using the same programs and year range.

8.0.2 Hardware and runtime environment

All pipeline stages were executed on the machine described in Table 1. The GPU was used exclusively by the local embedding model (*paraphrase-xlm-r-multilingual-v1* via *sentence-transformers*). All other pipeline stages, including GROBID extraction and Milvus vector storage, ran on the CPU.

Table 1: Hardware used for the experiment

Component	Specification
CPU	AMD Ryzen 5 7600 (6-core, 12-thread)
GPU	NVIDIA GeForce RTX 4070 Ti Super (16 GB VRAM)
RAM	32 GB DDR5
OS	Windows 11

8.0.3 Reproducibility and open science measures

The pipeline and all analysis scripts are released publicly on GitHub. Dependencies are pinned in a `requirements.txt` file, and the supported Python version is explicitly stated in the project documentation. Any researcher with access to the same infrastructure can reproduce the full pipeline run given an equivalent input corpus.

Reproducibility is further supported by two caching layers embedded in the architecture. First, GROBID structured-text extractions are cached on disk under `cache/tei/`, keyed by DOI. A second run over the same corpus therefore does not re-extract already-processed references; the pipeline reads the cached TEI XML directly. Second, embedding vectors are stored in Milvus and are keyed by DOI and model name. Re-running verification for a reference that was already embedded produces identical similarity scores without recomputation. Together these two caches mean that results are stable across runs and that partial failures, such as a network timeout during source retrieval, can be resumed without restarting from scratch.

The classification thresholds used throughout are explicit constants in the codebase: a cosine similarity of at least 0.62 is required for the *supported* label, and at least 0.52 for the *related* label. All citations below 0.52 receive the *no_support* label. Both values are configurable via command-line arguments, so any downstream study can adjust them and regenerate all figures with the provided analysis scripts.

8.0.4 Reference corpus characteristics

Figure 3 shows the distribution of bibliography sizes across the four faculties.

Bibliography sizes vary considerably both within and across faculties. Some theses contain as few as ten references while others exceed a hundred, reflecting the breadth of topics covered within each program. This variation matters for the

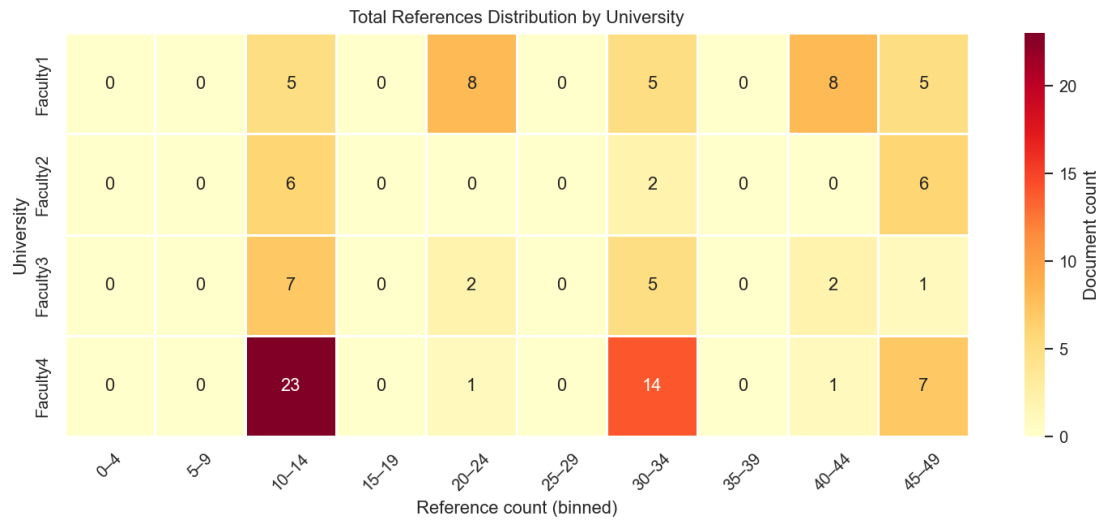


Figure 3: Distribution of bibliography sizes by faculty. Each cell shows the number of theses whose reference list falls within the corresponding bin.

pipeline: a thesis with a small, tightly focused bibliography gives the verifier fewer but more targeted citation pairs to check, whereas a thesis with a large bibliography generates more pairs and places greater demand on source retrieval.

Figure 4 shows the mean DOI coverage per faculty. The fraction of references for which a valid DOI was detected in the bibliography entry.

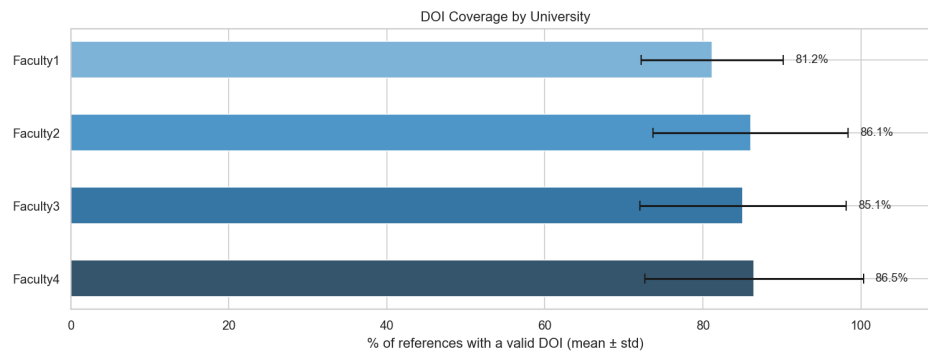


Figure 4: Mean DOI coverage by faculty (mean $\pm$ std). A valid DOI is required for reliable source retrieval via Crossref.

DOI coverage is consistently high across all four faculties, ranging from roughly [PLACEHOLDER: 81]% to [PLACEHOLDER: 87]%. This indicates that the majority of cited works in the corpus are indexed in digital databases and carry a

persistent identifier, which is a prerequisite for reliable source retrieval. The remaining [PLACEHOLDER: 13–19]% without a valid DOI are handled by a full-text search fallback, though this produces less reliable matches and a lower retrieval success rate.

8.0.5 Source retrieval results

Before semantic verification can take place, the pipeline must obtain the text of each cited reference. Figure 5 shows the retrieval outcome breakdown across all four faculties.

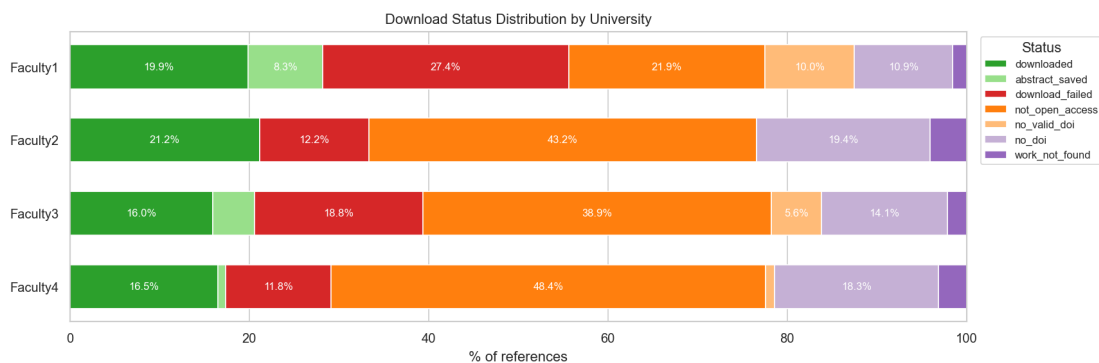


Figure 5: Source retrieval outcome distribution by faculty. Each bar shows the proportion of references falling into each retrieval category.

Across the corpus, roughly [PLACEHOLDER: 16–21]% of references yielded a downloadable full-text PDF. An additional [PLACEHOLDER: 6–9]% produced at least an abstract, which serves as a fallback evidence source. The largest single barrier was open-access restrictions: between [PLACEHOLDER: 38]% and [PLACEHOLDER: 48]% of references per faculty fell into the *not_open_access* category, meaning a DOI was resolved but the publisher did not provide free access to the full text. A further [PLACEHOLDER: 12–27]% resulted in a download failure despite an available DOI, most commonly due to dead links or API rate limits at the time of retrieval.

These retrieval rates directly determine what fraction of the corpus can be semantically verified. Citations whose source material was unavailable are excluded from the similarity analysis, so the effective verified fraction varies by faculty depending on the open-access profile of the cited literature.

8.0.6 Similarity analysis results

Figure 6 shows the overall distribution of best cosine similarity scores across all verified citations in the corpus.

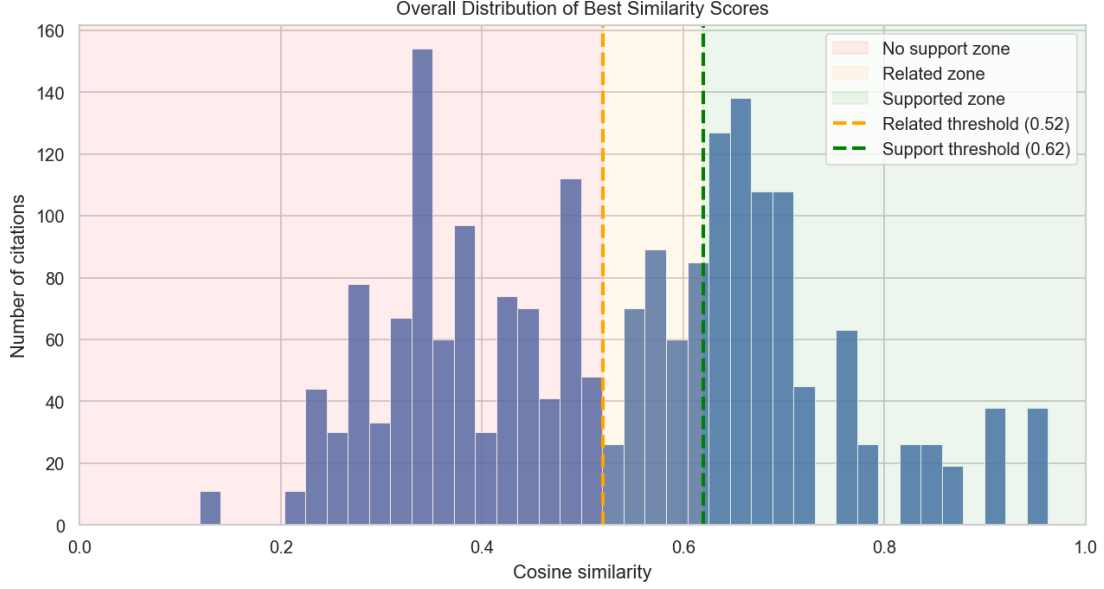


Figure 6: Distribution of best cosine similarity scores across all verified citations. Vertical dashed lines indicate the *related* threshold (0.52) and the *supported* threshold (0.62). Coloured bands mark the corresponding label zones.

The distribution is bimodal. A substantial share of citations falls in the 0.30–0.55 range, sitting below or near the *related* threshold, while a second concentration appears just above the *supported* threshold at 0.62. This shape suggests that the model can often distinguish between citations that are genuinely topically close to their source and those that are not, but it also reflects the difficulty of cross-lingual comparison when the citing text and the referenced source use different languages or domain-specific terminology.

Figure 7 breaks the same data down by faculty.

The share of *supported* citations varies across faculties, ranging from roughly [PLACEHOLDER: 29]% to [PLACEHOLDER: 45]%. The *no_support* category consistently accounts for the largest fraction at each faculty, between [PLACEHOLDER: 43]% and [PLACEHOLDER: 50]%. These differences should be interpreted with caution. The retrieval rate — and therefore the depth of available evidence — differs by faculty depending on the disciplinary mix of the cited liter-

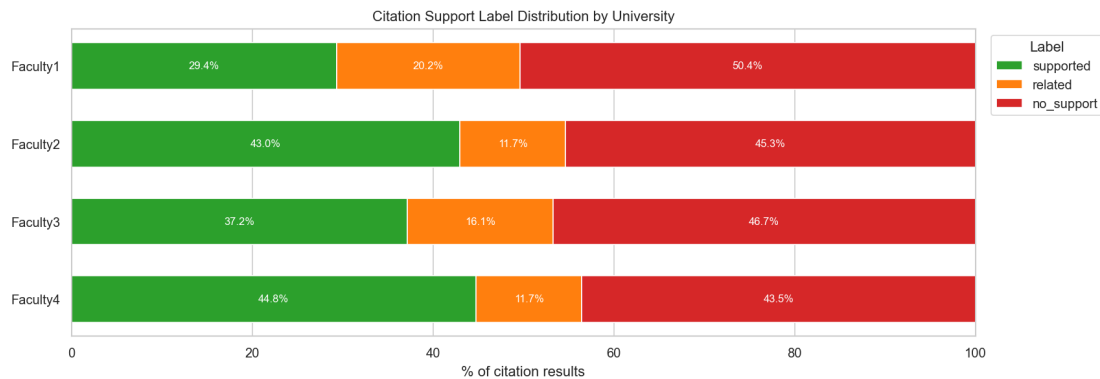


Figure 7: Citation support label distribution by faculty. Each bar shows the percentage of verified citations assigned the *supported*, *related*, or *no_support* label.

ature and the open-access availability of sources in those fields. A faculty whose students tend to cite closed-access journals will naturally show lower verified ratios and lower mean similarity scores, regardless of any differences in citation practice. No inference about the quality of academic oversight at any particular institution is intended or supportable from these figures alone.

Figure 8 provides complementary information about citation behaviour, showing how often individual references are cited within a thesis across the four faculties.

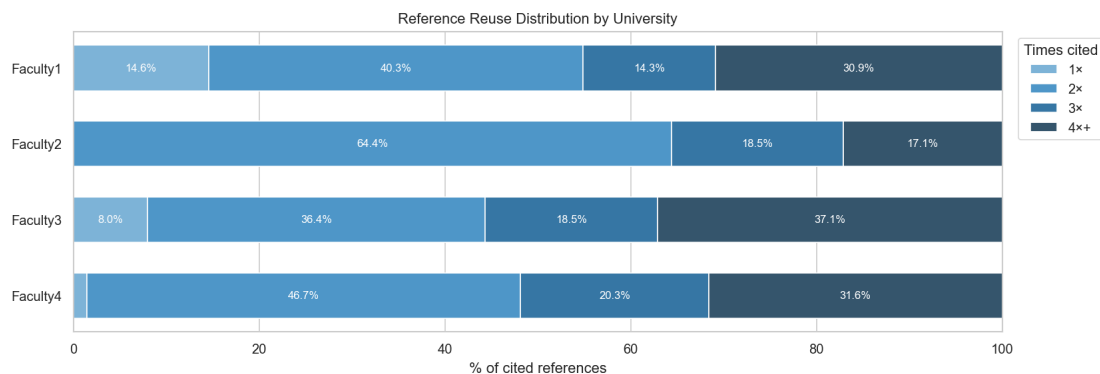


Figure 8: Reference reuse distribution by faculty. Each bar shows the proportion of cited references that appear exactly once, twice, three times, or four or more times within a given thesis.

Citation behaviour differs noticeably between faculties. In some, the majority of references are cited only once or twice, suggesting that each source plays a

narrowly scoped supporting role. In others, a substantial fraction of references appear four or more times, indicating that a smaller set of sources is drawn on repeatedly throughout the thesis. This variation is relevant when interpreting similarity results: a reference cited many times across diverse contexts is inherently harder to verify uniformly, since different citing sentences may target different parts of the same source.

8.0.7 Interpretation and limitations

The results presented here are indicative rather than conclusive. Two sources of imprecision accumulate across the pipeline. First, citation and reference detection is not perfect: inline citations may be missed if the PDF has unusual formatting, and bibliography parsing can fail for entries that deviate from the expected IEEE style. Second, source retrieval is constrained by open-access availability, meaning a significant portion of the corpus is verified against abstracts alone rather than full papers, which reduces the chance of finding a matching passage.

The most important interpretive caution concerns the *no_support* label. A low cosine similarity between a citing sentence and any retrieved passage from the referenced source does not by itself indicate that the citation is misused or that the author has copied without attribution. The student may have drawn a broader or more general conclusion from the work than the sentence-level similarity score can capture. Paraphrase, cross-lingual reformulation, and high-level synthesis are all valid scholarly practices that tend to produce lower similarity scores than verbatim or near-verbatim copying. The role of the pipeline is therefore to flag citations for closer inspection, not to render a verdict. Any case of genuine concern identified by the system should be reviewed by a human assessor with access to both the original thesis and the cited source before any conclusion is drawn.

9 Sustainability and Environmental Impact

The long-term value of this project depends less on any single model and more on whether the whole solution can remain understandable, reproducible, and maintainable after the thesis is finished. Student software often loses value because it is tightly coupled to one environment, one dependency stack, or one undocumented workflow. In this work, sustainability is supported mainly by keeping the pipeline modular and by separating the problem into citation extraction, source retrieval, and semantic verification. These stages solve different tasks and communicate through explicit intermediate outputs, so a future maintainer can modify one part

without having to redesign the rest of the system. If a PDF parsing tool, retrieval source, or embedding model becomes obsolete, it can be replaced locally while preserving the overall workflow.

Reducing future maintenance also requires limiting hidden assumptions. The implementation relies on explicit configuration, versioned dependencies, and fixed similarity thresholds that are documented rather than learned implicitly during training. This makes the behaviour of the system easier to inspect and reproduce. External services such as GROBID, Crossref, and Milvus are accessed through well-defined interfaces, which lowers the cost of future migration if one of these services changes or is replaced. The caching strategy described earlier also contributes to sustainability. Once structured text or embeddings have been created, the pipeline can reuse them in later runs instead of repeating the same expensive work or depending again on external services that may later become unavailable.

Documentation and archiving are equally important. The code repository, setup instructions, and analysis scripts should remain archived together with the final thesis so that the project can still be reconstructed after several years. A sustainable archive should contain the exact runtime requirements, the expected service setup, example commands, and sample outputs of the generated protocol. At the same time, the thesis corpus itself should not be redistributed, because public availability of theses does not remove the privacy concerns connected with assembling them into a research dataset. For that reason, the sustainable path is to archive the method, the code, and the instructions for rebuilding an equivalent dataset, rather than the collected corpus itself.

The machine learning component was also chosen with sustainability in mind. The system uses a pretrained multilingual sentence embedding model and does not require training a new model from scratch. This significantly lowers the technical barrier for future use, because extending the project with new theses only requires running the same pipeline over additional documents. New data can therefore be incorporated incrementally without retraining. If a stronger multilingual encoder becomes available in the future, the model can be replaced at the configuration level and the similarity thresholds can be recalibrated on a validation sample. Such a replacement would still be much cheaper and simpler than designing and training a new domain-specific model. It also keeps the project open to future improvement without locking it to one model family.

This design choice matters from an environmental perspective as well. Training or repeatedly fine-tuning large multilingual models is computationally expensive and would create a recurring energy cost every time the dataset grows or the

model changes. By reusing pretrained models and caching extracted text and embeddings, the current solution avoids most of that cost. The main recurring computation is inference, and even that can be reduced because already processed material does not need to be analysed again unless the configuration changes in a meaningful way. If the tool is later deployed for real institutional use, a hosted service would probably be more sustainable than repeated local installations on many individual machines. Central deployment would make updates easier, reduce duplicated setup effort, and allow compute resources to be used only when verification is actually requested.

10 Employability

This project strengthened skills that are directly relevant to current IT and AI roles. It involved document parsing, information retrieval, multilingual NLP, vector search and usage of external APIs. These are practical tasks that appear in applied machine learning, search, and document-intelligence systems, so the thesis gave me experience that transfers well beyond the academic setting.

An important part of that experience came from working with Slovak and English texts in a setting where data quality and source availability were often imperfect. That forced me to think beyond ideal input and design a system that remains useful even when some parts fail or return uncertain results. At the same time, the project improved my ability to narrow a broad idea into a feasible scope, justify technical decisions, and explain the result clearly to other people. This combination of technical and communication skills strengthens my employability far more than solving an isolated programming task would.

11 Teamwork

Although the implementation was individual, the project developed through discussions with people from different backgrounds and through presenting the work at IITSRC. Those conversations were useful because they forced me to explain the solution in a practical way, not only as a technical pipeline. Instead of focusing only on models and similarity scores, I had to show what the tool would actually provide to a reviewer and where human judgment would still be necessary.

Teamwork was also important when defining the scope of the thesis. The original idea was a more general multilingual plagiarism detector, but during consultations

it became clear that citation verification was a more focused and realistic direction. Presenting the work and discussing it with others helped confirm that this narrower scope had clearer evaluation criteria and more practical value for academic review. My role in that process was to act as the technical expert, propose implementable solutions, and translate feedback into concrete design decisions.

12 Conclusions

This thesis showed that cross-lingual citation verification can be implemented as a practical support tool for academic review. The result is not a system that should replace a human evaluator, but a pipeline that can make review work faster and more transparent by identifying citation contexts, retrieving relevant sources, and showing passages that appear to support the cited claim. The main contribution of the work is therefore the design of an end-to-end workflow that helps reviewers focus their attention where it is most needed.

From a technical point of view, the thesis demonstrated that this problem can be addressed with a modular architecture built from existing tools and pretrained multilingual models. The system connects PDF parsing, reference matching, source retrieval, passage comparison, and protocol generation into one process. The evaluation on 1000 Slovak bachelor and master theses also showed that the approach is meaningful on a realistic corpus. At the same time, it became clear that source availability is one of the main limiting factors, because many references cannot be checked against full text even when the semantic model itself is usable.

The practical value of the project lies in supporting reviewers, librarians, and academic integrity staff with evidence rather than a single final score. In that sense, the work has broader relevance than this thesis alone. It addresses a gap left by traditional plagiarism detection systems, which are much stronger at detecting direct lexical overlap than at handling cross-lingual reformulation supported by formal citations. The same general approach could also be useful in other tasks where claims need to be connected with source passages across multilingual documents.

On a personal level, the project improved my ability to move from a broad idea to a narrower and more defensible technical solution. It required work across NLP, retrieval, software engineering, and evaluation, but also clear communication about uncertainty and practical limitations. That combination was one of the most valuable outcomes of the thesis and confirmed my interest in building language technologies that support real user workflows.

The most natural future direction is to improve the usability of the existing solution. If the tool is used by universities, it should gain a graphical or web-based interface with more feedback for the reviewer. In practice, an online deployment would likely be more useful than running the application locally. Before full verification starts, the system could show all detected citations, references, and extracted sentences so the reviewer can correct them. After retrieval, it should display the downloaded texts by name and allow the reviewer to reject incorrect results or add missing ones. These additions would make the tool more suitable for real institutional use while preserving the main contribution of this work.

13 Résumé

TBD

Literature

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14 Scientific Part of the Thesis

TBD

Appendices

A Appendices

B Work Plan

C Technical Documentation

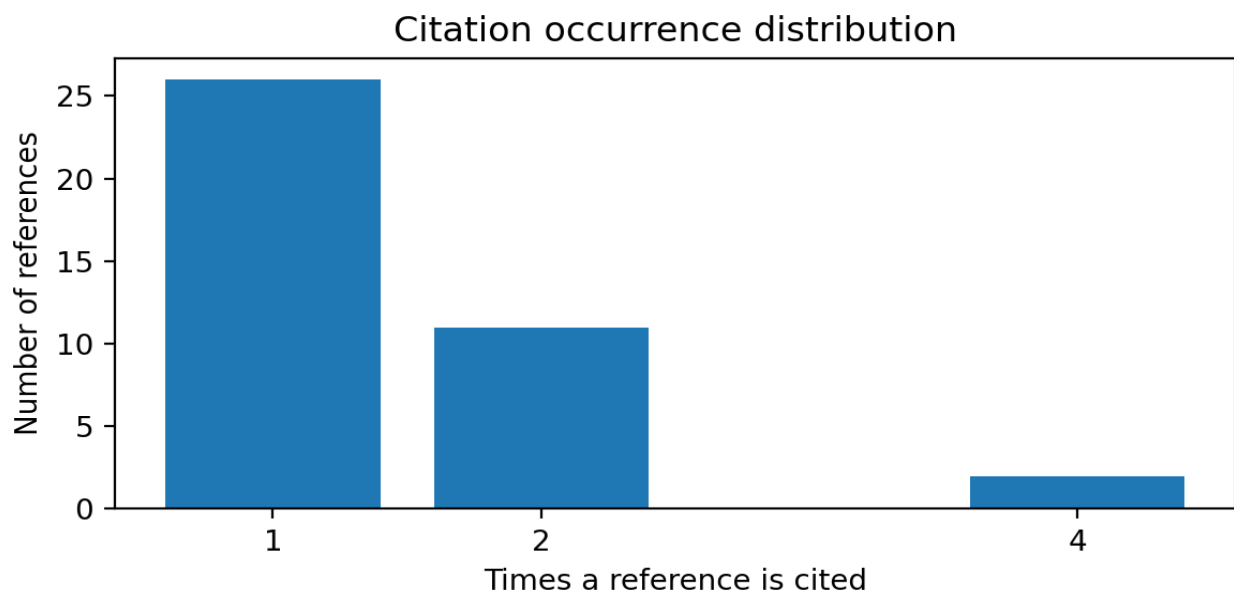
D Contents of the Digital Medium

Citation Verification Protocol

Source PDF: D:\Uni\bp-text\build\main.pdf

Paper statistics

Total references	39
Cited references	39
Unused references	0
Total citations (in-text)	56
Unmatched citation entries	0
Unmatched citation labels	0

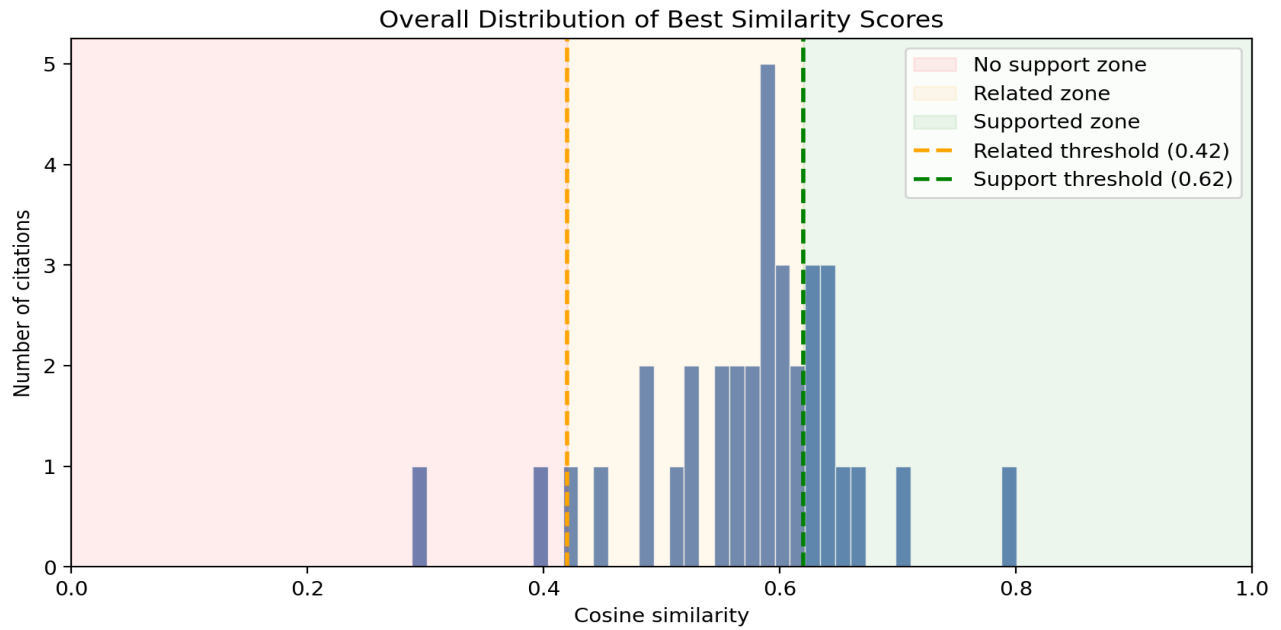


Summary

References (links): 39

Verified citing sentences: 35

Labels: supported=10, related=23, no_support=2



Reference [1]

Occurrences in work: 1

Resolved DOI: 10.24097/wolfram.96543.data

Resolved title: 1911 Encyclopedia Britannica

Download status: download_failed

Raw reference: [1] Encyclopedia Britannica 2025. 2025. url: <https://www.britannica.com/topic/plagiarism>.

Verification status: no_source_text

Reference [2]

Occurrences in work: 1

Resolved DOI: 10.1109/incacct57535.2023.10141785

Resolved title: A Review on diverse algorithms used in the context of Plagiarism Detection

Download status: download_failed

Raw reference: [2] Anchal Pokharana and Urvashi Garg. "A Review on diverse algorithms used in the context of Plagiarism Detection". In: 2023 International Conference on Advancement in Computation & Computer Technologies (InCACCT). 2023, pp. 1–6. doi: 10.1109/InCACCT57535.2023.10141785.

Verification status: no_source_text

Reference [3]

Occurrences in work: 1

Resolved DOI: 10.1016/j.urolonc.2010.09.013

Resolved title: The problem of plagiarism

Download status: downloaded

Downloaded PDF: pdfs\10.1016_j.urolonc.2010.09.013.pdf

Raw reference: [3] Melissa S. Anderson and Nicholas H. Steneck. "The problem of plagiarism". In: Urologic Oncology: Seminars and Original Investigations 29.1 (2011), pp. 90–94. issn: 1078-1439. doi: <https://doi.org/10.1016/j.urolonc.2010.09.013>. url: <https://www.sciencedirect.com/science/article/pii/S107814391000270X>.

Citing sentences and best evidence

1. Label: related **sim:** 0.587 **hint:** unclear **zone:** body

Citing: Anderson and Steneck[3] emphasize that the ethical problem begins when borrowed ideas are reused without proper acknowledgment, not when researchers merely learn from prior work.

- **Match 1** (sim 0.587) [body/paragraph] Introduction: Researchers' careers are built on their ideas and their contributions to the research record. It should be no surprise, then, that plagiarism is viewed by the research community as a serious violation of the norms of research. It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original id...

- **Match 2** (sim 0.559) [body/section] Introduction: Researchers' careers are built on their ideas and their contributions to the research record. It should be no surprise, then, that plagiarism is viewed by the research community as a serious violation of the norms of research. It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original id...

- **Match 3** (sim 0.530) [body/sentence] Introduction: It constitutes tampering with the system by which researchers' work is recognized and rewarded, and it is a personal affront and act of disrespect to the individual who wrote the original words or expressed the original idea.

Reference [4]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [4] Zákon ■. 131/2002 Z. z. Zbierka zákonov Slovenskej republiky. 2002. url: <https://www.slov-lex.sk/ezbierky/pravne-predpisy/SK/ZZ/2002/131/20230101>.

Verification status: no_source_text

Reference [5]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [5] Zákon ■. 185/2015 Z. z. Zbierka zákonov Slovenskej republiky. 2015. url: <https://www.slov-lex.sk/ezbierky/pravne-predpisy/SK/ZZ/2015/185/>.

Verification status: no_source_text

Reference [6]

Occurrences in work: 2

Resolved DOI: 10.1007/s10579-009-9114-z

Resolved title: Cross-language plagiarism detection

Download status: abstract_saved

Raw reference: [6] Martin Potthast et al. "Cross-language plagiarism detection". In: Language Resources and Evaluation 45.1 (2011), pp. 45–62. issn: 1574-0218. doi: 10. 1007/s10579-009-9114-z. url: <https://doi.org/10.1007/s10579-0099114-z>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.672 **hint:** mostly_abstract **zone:** abstract

Citing: Cross-lingual plagiarism detection studies cases in which a document in one language reuses material from a source in another language, either through direct translation or paraphrased translation[6].

- **Match 1** (sim 0.672) [abstract/sentence] abstract: Cross-language plagiarism detection deals with the automatic identification and extraction of plagiarism in a multilingual setting.
- **Match 2** (sim 0.617) [abstract/paragraph] abstract: Cross-language plagiarism detection deals with the automatic identification and extraction of plagiarism in a multilingual setting. In this setting, a suspicious document is given, and the task is to retrieve all sections from the document that originate from a large, multilingual document collection. Our contributions in this field are as follows: (1) a comprehensive retrieval process for cross-language plagiarism detection is introduced, highl...
- **Match 3** (sim 0.475) [abstract/sentence] abstract: Our contributions in this field are as follows: (1) a comprehensive retrieval process for cross-language plagiarism detection is introduced, highlighting the differences to monolingual plagiarism detection, (2) state-of-the-art solutions for two important subtasks are reviewed, (3) retrieval models for the assessment of cross-language similarity are surveyed, and, (4) the three models CL-CNG, CL-ESA and CL-ASA are compared.

2. Label: related **sim:** 0.488 **hint:** weak_abstract **zone:** abstract

Citing: This kind of misuse is harder to detect than literal copying because the wording changes while the argumentative role of the source remains central[6].

- **Match 1** (sim 0.488) [abstract/sentence] abstract: Cross-language plagiarism detection deals with the automatic identification and extraction of plagiarism in a multilingual setting.
- **Match 2** (sim 0.446) [abstract/paragraph] abstract: Cross-language plagiarism detection deals with the automatic identification and extraction of plagiarism in a multilingual setting. In this setting, a suspicious document is given, and the task is to retrieve all sections from the document that originate from a large, multilingual document collection. Our contributions in this field are as follows: (1) a comprehensive retrieval process for cross-language plagiarism detection is introduced, highl...
- **Match 3** (sim 0.383) [abstract/sentence] abstract: In this setting, a suspicious document is given, and the task is to retrieve all sections from the document that originate from a large, multilingual document collection.

Reference [7]

Occurrences in work: 4

Resolved DOI: 10.1021/acs.chas.4c00097.s001

Resolved title: Can Large Language Models (LLMs) Act as Virtual Safety Officers?

Download status: download_failed

Raw reference: [7] Ahmad Zeeb and Philip Olsson. Citation Evaluation Using Large Language Models (LLMs) : Can LLMs evaluate citations in scholarly documents? An experimental study on ChatGPT. 2024.

Verification status: no_source_text

Reference [8]

Occurrences in work: 2

Resolved DOI: 10.18653/v1/N19-1423.url:<https://aclanthology.org/N19-1423/>

Download status: no_valid_doi

Raw reference: [8] Jacob Devlin et al. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota, June 2019. doi: 10.18653/v1/N19-1423. url: <https://aclanthology.org/N19-1423/>.

Verification status: no_source_text

Reference [9]

Occurrences in work: 2

Resolved DOI: 10.18653/v1/2020.acl-main.747

Resolved title: Unsupervised Cross-lingual Representation Learning at Scale

Download status: downloaded

Downloaded PDF: pdfs\10.18653_v1_2020.acl-main.747.pdf

Raw reference: [9] Alexis Conneau et al. “Unsupervised Cross-lingual Representation Learning at Scale”. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Ed. by Dan Jurafsky et al. Online: Association for Computational Linguistics, July 2020, pp. 8440–8451. doi: 10.18653/v1/2020.acl-main.747. url: <https://aclanthology.org/2020.aclmain.747/>. Literature

Citing sentences and best evidence

1. Label: related **sim:** 0.587 **hint:** unclear **zone:** body

Citing: Research in cross-lingual plagiarism detection has moved from string matching and translation resources toward embedding-based approaches built on transformer architectures such as mBERT[8] and XLM-R[9].

- **Match 1** (sim 0.587) [body/paragraph] Introduction: Multilingual masked language models (MLM) like mBERT (Devlin et al., 2018) and XLM (Lample and Conneau, 2019) have pushed the state-of-the-art on cross-lingual understanding tasks by jointly pretraining large Transformer models (Vaswani et al., 2017) on many languages. These models allow for effective cross-lingual transfer, as seen in a number of benchmarks including cross-lingual natural language inference (Bowman et al., 2015; Williams et al., ...)

- **Match 2** (sim 0.568) [body/sentence] Introduction: Multilingual masked language models (MLM) like mBERT (Devlin et al., 2018) and XLM (Lample and Conneau, 2019) have pushed the state-of-the-art on cross-lingual understanding tasks by jointly pretraining large Transformer models (Vaswani et al., 2017) on many languages.

- **Match 3** (sim 0.558) [body/sentence] Representation Learning for Low-resource Languages: This confirms our assumption that mBERT and XLM-100 rely heavily on cross-lingual transfer but do not model the low-resource languages as well as XLM-R. Specifically, in the translate-train-all setting, we observe that the biggest gains for XLM models trained on CC, compared to their Wikipedia counterparts, are on low-resource languages; 7% and 4.8% improvement on Swahili and Urdu respectively.

2. Label: supported **sim:** 0.643 **hint:** body **zone:** body

Citing: XLM-R[9] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[20]-style training improvements such as dynamic masking and larger batches.

- **Match 1** (sim 0.643) [body/sentence] Related Work: Inspired by RoBERTa, we show that mBERT and XLM are undertuned, and that simple improvements in the learning procedure of unsupervised MLM leads to much better performance.

- **Match 2** (sim 0.589) [body/sentence] Model: While XLM-R is among the largest models partly due to its large embedding layer, it has a similar number of parameters than XLM-100, and remains significantly smaller than recently introduced Transformer models for multilingual MT and transfer learning.

- **Match 3** (sim 0.583) [body/section] Introduction: The goal of this paper is to improve cross-lingual language understanding (XLU), by carefully studying the effects of training unsupervised crosslingual representations at a very large scale. We present XLM-R a transformer-based multilingual masked language model pre-trained on text in 100 languages, which obtains state-of-the-art performance on cross-lingual classification, sequence labeling and question answering. Multilingual masked language ...

Reference [10]

Occurrences in work: 2

Resolved DOI: 10.48550/arxiv.2402.14268

Resolved title: Can Large Language Models Detect Misinformation in Scientific News Reporting?

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2402.14268.pdf

Raw reference: [10] Yupeng Cao et al. Can Large Language Models Detect Misinformation in Scientific News Reporting? 2025. arXiv: 2402.14268 [cs.CL]. url: <https://arxiv.org/abs/2402.14268>.

Citing sentences and best evidence

1. Label: related **sim:** 0.600 **hint:** unclear **zone:** body

Citing: At the same time, citationverification research is starting to explore claim-evidence modelling[10], full-text evidence retrieval[11], and large language models for assessing whether a cited claim is supported by a source passage[7].

- **Match 1** (sim 0.600) [body/sentence] Fact-Checking: Automatic fact-checking, which assesses the truthfulness of claims made in the text [15,40], has been extensively studied across various domains, including common knowledge verification [38], political topics [49], COVID-19 [31], E-commercial [60], biology [10].

- **Match 2** (sim 0.592) [body/section] Fact-Checking: Automatic fact-checking, which assesses the truthfulness of claims made in the text [15,40], has been extensively studied across various domains, including common knowledge verification [38], political topics [49], COVID-19 [31], E-commercial [60], biology [10]. When thinking about scientific misrepresentation in popular media, it is natural to think of the veracity of scientific findings. It is therefore possible to cast the scientific misinfor...

- **Match 3** (sim 0.592) [body/paragraph] Fact-Checking: Automatic fact-checking, which assesses the truthfulness of claims made in the text [15,40], has been extensively studied across various domains, including common knowledge verification [38], political topics [49], COVID-19 [31], E-commercial [60], biology [10]. When thinking about scientific misrepresentation in popular media, it is natural to think of the veracity of scientific findings. It is therefore possible to cast the scientific misinfor...

2. Label: supported **sim:** 0.622 **hint:** body **zone:** body

Citing: Large language models are now also being tested in neighbouring tasks such as scientific-misinformation detection[10], but these approaches still raise questions of reproducibility, cost, and consistency.

- **Match 1** (sim 0.622) [body/paragraph] INTRODUCTION: Several websites are maintained by science reporters (Health News Review 3) and scientists (Science Feedbackfoot_1) to track scientific misinformation in the media. Although such manual debunking is important, the sheer volume of scientific news can make this task unscalable. Natural language processing (NLP) based approaches have consequently started to emerge to deal with this problem. These methods typically involve language analysis, like ...

- **Match 2** (sim 0.607) [body/paragraph] INTRODUCTION: While these works have laid the foundation to address this problem, the area is still in its nascence. Several challenges remain unaddressed: 1) there are no existing taxonomies to define dimensions of the scientific validity of scientific news articles that can be used in automated methods to detect misinformation in scientific news; 2) all existing datasets for scientific fact-checking relies on explicit claim generation from the news articles...

- **Match 3** (sim 0.595) [body/section] RELATED WORK: As mentioned earlier, scientific misinformation detection is still in its nascence and while related to misinformation detection in general, it is a harder problem since the language characteristics of the scientific communication is different from the formal format of scientific publications. The problem of scientific misinformation is related to two other concepts in NLP, including: 1) fact-checking (claim verification) 2) scientific language ...

Reference [11]

Occurrences in work: 4

Resolved DOI: 10.48550/arxiv.2511.16198

Resolved title: SemanticCite: Citation Verification with AI-Powered Full-Text Analysis and Evidence-Based Reasoning

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2511.16198.pdf

Raw reference: [11] Sebastian Haan. SemanticCite: Citation Verification with AI-Powered FullText Analysis and Evidence-Based Reasoning. 2025. arXiv: 2511 . 16198 [cs.CL]. url: <https://arxiv.org/abs/2511.16198>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.644 **hint:** body **zone:** body

Citing: At the same time, citationverification research is starting to explore claim-evidence modelling[10], full-text evidence retrieval[11], and large language models for assessing whether a cited claim is supported by a source passage[7].

- **Match 1** (sim 0.644) [body/section] Citation Analysis and Verification: Traditional citation analysis has primarily focused on bibliometric approaches using citation counts as indicators of research impact, failing to capture semantic relationships between citing statements and referenced sources. The field of automated fact-checking has provided important foundations for citation verification. Wadden et al. (2020) [8] introduced scientific claim verification with their SciFact dataset comprising 1.4K expert-written...

- **Match 2** (sim 0.644) [body/paragraph] Citation Analysis and Verification: Traditional citation analysis has primarily focused on bibliometric approaches using citation counts as indicators of research impact, failing to capture semantic relationships between citing statements and referenced sources. The field of automated fact-checking has provided important foundations for citation verification. Wadden et al. (2020) [8] introduced scientific claim verification with their SciFact dataset comprising 1.4K expert-written...

- **Match 3** (sim 0.602) [body/sentence] Citation Analysis and Verification: Some existing platforms analyse citation usage within citing documents, extracting surrounding context and classifying whether the citing statement supports or contradicts the reference generally [11].

2. Label: related **sim:** 0.587 **hint:** unclear **zone:** body

Citing: An even more direct systems-level point of comparison is Haan's[11] SemanticCite, which proposes citation verification through AI-powered full-text analysis and evidence-based reasoning.

- **Match 1** (sim 0.587) [body/paragraph] Implications for Research Quality and Broader Applications: The demonstrated effectiveness of SemanticCite has broader implications for research quality and scholarly communication. Automated citation verification tools can significantly enhance peer review efficiency by identifying problematic citations before human review, allowing reviewers to focus on higher-level scientific assessment rather than time-consuming fact-checking.

- **Match 2** (sim 0.578) [body/section] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. Sema-ticCite bridges the gap between automated...

- **Match 3** (sim 0.578) [body/paragraph] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. Sema-ticCite bridges the gap between automated...

3. Label: supported **sim:** 0.628 **hint:** body **zone:** body

Citing: The system must work with full-text documents, identify citations, retrieve and analyse source material, and produce evidence that can justify the verification result[11].

- **Match 1** (sim 0.628) [body/sentence] Research Objectives: First, we develop an automated full-text citation verification system that transcends surface-level bibliographic checks to perform deep semantic analysis of citation content against complete source documents.
- **Match 2** (sim 0.613) [body/sentence] System Architecture: The system processes reference documents through text extraction and chunking, whilst citations undergo preprocessing to remove attribution markers and extract core factual claims.
- **Match 3** (sim 0.607) [body/paragraph] System Architecture: The citation verification system takes three inputs: citation text, reference document, and optional metadata. Then it produces a classification label, confidence score, supporting evidence, and detailed reasoning. The system processes reference documents through text extraction and chunking, whilst citations undergo preprocessing to remove attribution markers and extract core factual claims. Document processing begins with text extraction using...

4. Label: related **sim:** 0.619 **hint:** unclear **zone:** abstract

Citing: In this sense, the design is close to SemanticCite, which also approaches citation verification as an evidence-based process[11].

- **Match 1** (sim 0.619) [abstract/sentence] abstract: SemanticCite addresses critical challenges in research integrity through scalable citation verification, streamlined peer review, and quality control for AI-generated content, providing an open-source foundation for maintaining citation accuracy at scale.
- **Match 2** (sim 0.607) [body/sentence] Conclusion: This paper introduces SemanticCite, an AI-driven framework for automated citation verification that addresses fundamental limitations in existing approaches.
- **Match 3** (sim 0.600) [body/section] Contributions: This paper introduces SemanticCite, the first comprehensive AI-powered framework for automated fulltext citation verification. Unlike existing approaches that rely on abstract-level analysis or binary classification schemes, our system performs deep semantic analysis of complete source documents using a 4-class taxonomy that captures the nuanced relationships between citations and their references. SemanticCite bridges the gap between automated...

Reference [12]

Occurrences in work: 2

Resolved DOI: 10.48550/arxiv.1801.06323

Resolved title: Plagiarism: Taxonomy, Tools and Detection Techniques

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.1801.06323.pdf

Raw reference: [12] Hussain A Chowdhury and Dhruba K Bhattacharyya. Plagiarism: Taxonomy, Tools and Detection Techniques. 2018. arXiv: 1801.06323 [cs.IR]. url: <https://arxiv.org/abs/1801.06323>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.801 **hint:** body **zone:** body

Citing: A widely cited taxonomy distinguishes deliberate copy-paste plagiarism, simple paraphrasing, patchwork paraphrasing, selfplagiarism, illegitimate-source plagiarism, retweet plagiarism, metaphor plagiarism, and idea plagiarism[12].

- **Match 1** (sim 0.801) [body/paragraph] Plagiarism and Its Types: Based on characteristics, plagiarism can also be categorized into literal and intelligent plagiarism. Literal plagiarism consists of copy-paste/clone, paraphrasing, self/recycled, and retweet plagiarism. The other form of plagiarism can be considered as intelligent type of plagiarism. In general, intrinsic plagiarism detection methods can detect paraphrasing, idea, and mosaic textual plagiarism sub-types [4], whereas, external plagiarism detecti...

- **Match 2** (sim 0.794) [body/sentence] Plagiarism and Its Types: Literal plagiarism consists of copy-paste/clone, paraphrasing, self/recycled, and retweet plagiarism.

- **Match 3** (sim 0.747) [body/sentence] Plagiarism and Its Types: In general, intrinsic plagiarism detection methods can detect paraphrasing, idea, and mosaic textual plagiarism sub-types [4], whereas, external plagiarism detection methods can detect clone, metaphor, retweet and possibly (with a low probability or none) self plagiarisms and error 404 [4].

2. Label: supported **sim:** 0.651 **hint:** body **zone:** abstract

Citing: In computer science, the same paper also highlights source-code plagiarism categories such as local manipulation, structure reordering, whitespace-only disguise, and even language switching, all of which show that superficial modifications are often used to evade detection[12].

- **Match 1** (sim 0.651) [abstract/sentence] abstract: This paper highlights few promising methods for plagiarism detection based on machine learning techniques.

- **Match 2** (sim 0.638) [body/sentence] Issues and Challenges: Most prominent methods have been able to address the major issues related to (i) salient syntactic and semantic feature extraction, (ii) handling of both monolingual and cross-lingual plagiarism detection, and (iii) detecting plagiarism in both text data and program source code with or without using references.

- **Match 3** (sim 0.624) [body/sentence] ithenticate [51][46]: This plagiarism detection system is not able to find simple copy-paste plagiarism but also can detect moderately disguised borrowing (word/phrase reordering, substitution of some words with synonyms [52].

Reference [13]

Occurrences in work: 2

Resolved DOI: 10.1016/j.ipm.2016.04.006

Resolved title: Candidate document retrieval for cross-lingual plagiarism detection using two-level proximity information

Download status: downloaded

Downloaded PDF: pdfs\10.1016_j.ipm.2016.04.006.pdf

Raw reference: [13] Nava Ehsan and Azadeh Shakery. "Candidate document retrieval for crosslingual plagiarism detection using two-level proximity information". In: Information Processing & Management 52.6 (2016), pp. 1004–1017. issn: 03064573. doi: <https://doi.org/10.1016/j.ipm.2016.04.006>. url: <https://www.sciencedirect.com/science/article/pii/S0306457316300784>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.708 **hint:** body **zone:** body

Citing: Cross-lingual plagiarism itself is usually defined as reuse in which a suspicious document in one language originates from a document in another language, either through exact translation or paraphrased translation[13].

- **Match 1** (sim 0.708) [body/sentence] Introduction: In automatic cross-language plagiarism detection, the task is to retrieve plagiarized text written in language L that has originated from another document in a language other than L. With the rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools, cross-language plagiarism has become a serious problem and its detection requires more attention.

- **Match 2** (sim 0.686) [body/section] Introduction: Plagiarism refers to unauthorised use of text, code and ideas (Potthast, Barrón-Cedeño, Stein, & Rosso, 2011). In automatic cross-language plagiarism detection, the task is to retrieve plagiarized text written in language L that has originated from another document in a language other than L. With the rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools...

- **Match 3** (sim 0.686) [body/paragraph] Introduction: Plagiarism refers to unauthorised use of text, code and ideas (Potthast, Barrón-Cedeño, Stein, & Rosso, 2011). In automatic cross-language plagiarism detection, the task is to retrieve plagiarized text written in language L that has originated from another document in a language other than L. With the rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools...

2. Label: related **sim:** 0.443 **hint:** weak_abstract **zone:** abstract

Citing: Prior work From Ehsan et. al[13] distinguishes syntax-based, dictionary-based, parallel-corpusbased, comparable-corpus-based, and multilingual semantic-network-based strategies for comparing such texts.

- **Match 1** (sim 0.443) [abstract/paragraph] abstract: The rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools have caused cross-lingual plagiarism detection research area to receive increasing attention in recent years. The task of cross-language plagiarism detection entails two main steps: candidate retrieval and assessing pairwise document similarity. In this paper we examine candidate retrieval, where th...

- **Match 2** (sim 0.431) [abstract/sentence] abstract: The rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools have caused cross-lingual plagiarism detection research area to receive increasing attention in recent years.

- **Match 3** (sim 0.411) [body/sentence] Introduction: In automatic cross-language plagiarism detection, the task is to retrieve plagiarized text written in language L that has originated from another document in a language other than L. With the rapid growth of documents in different languages, the increased accessibility of electronic documents, and the availability of translation tools, cross-language plagiarism has become a serious problem and its detection requires more attention.

Reference [14]

Occurrences in work: 2

Resolved DOI: 10.1145/3345317

Resolved title: Academic Plagiarism Detection

Download status: abstract_saved

Raw reference: [14] Tomáš Foltýnek, Norman Meuschke, and Bela Gipp. “Academic Plagiarism Detection: A Systematic Literature Review”. In: ACM Comput. Surv. 52.6 (Oct. 2019). issn: 0360-0300. doi: 10.1145/3345317. url: <https://doi.org/10.1145/3345317>.

Citing sentences and best evidence

1. Label: related **sim:** 0.597 **hint:** weak_abstract **zone:** abstract

Citing: Foltýnek et al.[14] note that “even in the best case, i.e., if the plagiarism is discovered, reviewing and punishing plagiarized research papers and grant applications still causes a high effort for the reviewers, affected institutions, and funding agencies”.

- **Match 1** (sim 0.597) [abstract/sentence] abstract: We show that academic plagiarism detection is a highly active research field.

- **Match 2** (sim 0.586) [abstract/paragraph] abstract: This article summarizes the research on computational methods to detect academic plagiarism by systematically reviewing 239 research papers published between 2013 and 2018. To structure the presentation of the research contributions, we propose novel technically oriented typologies for plagiarism prevention and detection efforts, the forms of academic plagiarism, and computational plagiarism detection methods. We show that academic plagiarism de...

- **Match 3** (sim 0.533) [abstract/sentence] abstract: To structure the presentation of the research contributions, we propose novel technically oriented typologies for plagiarism prevention and detection efforts, the forms of academic plagiarism, and computational plagiarism detection methods.

2. Label: related **sim:** 0.520 **hint:** weak_abstract **zone:** abstract

Citing: If it remains undiscovered, plagiarists may gain academic credit, funding, or career advancement based on misappropriated work[14].

- **Match 1** (sim 0.520) [abstract/sentence] abstract: We show that academic plagiarism detection is a highly active research field.

- **Match 2** (sim 0.474) [abstract/sentence] abstract: To structure the presentation of the research contributions, we propose novel technically oriented typologies for plagiarism prevention and detection efforts, the forms of academic plagiarism, and computational plagiarism detection methods.

- **Match 3** (sim 0.447) [abstract/sentence] abstract: Over the period we review, the field has seen major advances regarding the automated detection of strongly obfuscated and thus hard-to-identify forms of academic plagiarism.

Reference [15]

Occurrences in work: 1

Resolved DOI: 10.1177/0165551518787696

Resolved title: Cross-lingual text alignment for fine-grained plagiarism detection

Download status: abstract_saved

Raw reference: [15] Nava Ehsan, Azadeh Shakery, and Frank Wm Tompa. "Cross-lingual text alignment for fine-grained plagiarism detection". In: J. Inf. Sci. 45.4 (Aug. 2019), 443–459. issn: 0165-5515. doi: 10.1177/0165551518787696. url: <https://doi.org/10.1177/0165551518787696>.

Citing sentences and best evidence

1. Label: related **sim:** 0.579 **hint:** weak_abstract **zone:** abstract

Citing: Early work relied on dictionary-based translations[15] and syntax-based approaches[16], but these tend to be computationally expensive and limited by available resources.

- **Match 1** (sim 0.579) [abstract/sentence] abstract: In both steps, our approach uses a dictionary to obtain translations of individual terms instead of using a machine translation system to convert longer passages from one language to another.
- **Match 2** (sim 0.552) [abstract/sentence] abstract: Fast and easy access to a wide range of documents in various languages, in conjunction with the wide availability of translation and editing tools, has led to the need to develop effective tools for detecting cross-lingual plagiarism.
- **Match 3** (sim 0.435) [abstract/paragraph] abstract: Fast and easy access to a wide range of documents in various languages, in conjunction with the wide availability of translation and editing tools, has led to the need to develop effective tools for detecting cross-lingual plagiarism. Given a suspicious document, cross-lingual plagiarism detection comprises two main subtasks: retrieving documents that are candidate sources for that document and analysing those candidates one by one to determine ...

Reference [16]

Occurrences in work: 1

Resolved DOI: 10.1016/j.eswa.2017.07.006

Resolved title: Text plagiarism classification using syntax based linguistic features

Download status: downloaded

Downloaded PDF: pdfs\10.1016_j.eswa.2017.07.006.pdf

Raw reference: [16] Vani K and Deepa Gupta. "Text plagiarism classification using syntax based linguistic features". In: Expert Systems with Applications 88 (2017), pp. 448– 464. issn: 0957-4174. doi: <https://doi.org/10.1016/j.eswa.2017.07.006>. url: <https://www.sciencedirect.com/science/article/pii/S095741741730475X>.

Citing sentences and best evidence

1. Label: related **sim:** 0.423 **hint:** unclear **zone:** body

Citing: Early work relied on dictionary-based translations[15] and syntax-based approaches[16], but these tend to be computationally expensive and limited by available resources.

- **Match 1** (sim 0.423) [body/sentence] Introduction: In general, plagiarism can range from simple copy-paste where the data is literally copied to the higher complexity levels where the text is manipulated by sentence restructurings, idea adoptions, translations etc.

- **Match 2** (sim 0.416) [body/sentence] Introduction: Existing surveys and studies report that the most of the available plagiarism detection tools could not detect structural changes and common paraphrasing imposed by the plagiarists (Marsh, 2004;Mozgovoy, Kakkonen, & Cosma, 2010;Weber-Wulff, 20 08; Williams, 20 02; Clough, 20 0 0).

- **Match 3** (sim 0.402) [abstract/sentence] abstract: The objective is to explore the potency of syntax based linguistic features extracted using shallow natural language processing techniques for plagiarism classification task.

Reference [17]

Occurrences in work: 1

Resolved DOI: 10.1109/conmedia46929.2019.8981821

Resolved title: String Matching based Plagiarism Detection for Document in Bahasa Indonesia

Download status: download_failed

Raw reference: [17] Wayan Gede Suka Parwita, I Gusti Ayu Agung Diatri Indradewi, and I Nyoman Saputra Wahyu Wijaya. "String Matching based Plagiarism Detection for Document in Bahasa Indonesia". In: 2019 5th International Conference on New Media Studies (CONMEDIA). 2019, pp. 54–58. doi: 10.1109/CONMEDIA46929.2019.8981821.

Verification status: no_source_text

Reference [18]

Occurrences in work: 1

Resolved DOI: 10.3115/v1/d14-1153

Resolved title: Intrinsic Plagiarism Detection using N-gram Classes

Download status: downloaded

Downloaded PDF: pdfs\10.3115_v1_d14-1153.pdf

Raw reference: [18] Imene Bensalem, Paolo Rosso, and Salim Chikhi. "Intrinsic Plagiarism Detection using N-gram Classes". In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP). Ed. by Alessandro Moschitti, Bo Pang, and Walter Daelemans. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1459–1464. doi: 10.3115/v1/ D14-1153. url: <https://aclanthology.org/D14-1153/>. Literature

Citing sentences and best evidence

1. Label: related **sim:** 0.531 **hint:** unclear **zone:** body

Citing: Classical retrieval-oriented or lexical methods, including string matching[17] and n-gram[18] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[19].

- **Match 1** (sim 0.531) [body/paragraph] Introduction and Related Works: As opposed to the problem of authorship clustering, where the task is merely to attribute already defined fragments of a given document to different authors, the segmentation is a crucial and inevitable task in a real scenario of intrinsic plagiarism detection. Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

- **Match 2** (sim 0.518) [body/sentence] Introduction and Related Works: Indeed, a granular segmentation may lead to an undependable style analysis, and a coarse segmentation may prevent the identification of the short plagiarized texts.

- **Match 3** (sim 0.493) [body/paragraph] Introduction and Related Works: Unlike the authorship attribution and verification, where the examined text and the potential author text are separate (and hence their writing styles could be readily characterized and compared), these two parts are both merged in the same document with unknown boundaries. Furthermore, the plagiarized fragments in a suspicious document might stem from different authors, which renders the computational characterization of plagiarism difficult.

Reference [19]

Occurrences in work: 2

Resolved DOI: 10.3389/fcomp.2025.1504725

Resolved title: Plagiarism types and detection methods: a systematic survey of algorithms in text analysis

Download status: downloaded

Downloaded PDF: pdfs\10.3389_fcomp.2025.1504725.pdf

Raw reference: [19] Altynbek Amirzhanov, Cemil Turan, and Alfira Makhmutova. "Plagiarism types and detection methods: a systematic survey of algorithms in text analysis". In: *Frontiers in Computer Science* Volume 7 - 2025 (2025). issn: 2624-9898. doi: 10 . 3389 / fcomp . 2025 . 1504725. url: <https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1504725>.

Citing sentences and best evidence

1. Label: related **sim:** 0.515 **hint:** unclear **zone:** body

Citing: Classical retrieval-oriented or lexical methods, including string matching[17] and n-gram[18] techniques, are computationally simple and can work well for literal overlap, but they lose effectiveness under strong paraphrase or translation obfuscation[19].

- **Match 1** (sim 0.515) [body/sentence] Paraphrasing highly technical language: All of the above considerations serve to illustrate the reason why an operational definition of proper paraphrasing/plagiarism (i.e., how many consecutive words taken from the original constitutes plagiarism) is impractical, not to mention the fact that there are certain stock phrases, perhaps even entire sentences that occur with some frequency in unrelated journal articles (e.g., "the results obtained do not support the hypothesis").

- **Match 2** (sim 0.496) [body/sentence] Paraphrasing highly technical language: In sum, the reality is that in many instances, scientific prose and diction can be very difficult to paraphrase.

- **Match 3** (sim 0.496) [body/section] Redundancy, publication overlap and other forms of duplication: Although the prevalence of blatant duplicate publications varies across disciplines, its overall prevalence is relatively low (see Larivière & Gingras, 2010) and their impact on the integrity of science is likely minor, particularly in instances when the published papers are truly identical (i.e., same title, abstract, author list). However, other forms of duplication exist and these are often classified with terms such as redundant publication ...

2. Label: no_support **sim:** 0.401 **hint:** unclear **zone:** body

Citing: Amirzhanov et al.[19] note that this limitation has pushed the field away from symbolic and lexical strategies toward semantic representations better suited to multilingual comparison.

- **Match 1** (sim 0.401) [body/paragraph] Cultural-linguistic considerations of plagiarism and self-plagiarism: Admittedly, some of the evidence does suggest that many native English-speaking students (McGowan & Lightbody, 2008;Power, 2009;Roig, 1997;1999) and even some professionals (Julliard, 1994;Roig, 2001) are simply not familiar with the nuances of appropriate scholarship. In addition, the evidence also suggests that an important factor that leads writers to misappropriate long strings of text from sources is a combination of uniqueness of the termi...

- **Match 2** (sim 0.400) [body/sentence] Self-plagiarism within and across various other dissemination domains: In some disciplines, such as political science, the presentation of the same paper in multiple conferences has become a more common practice (Dometrius, 2008) and this development has been a source of concern for some in that discipline about possible skewed perceptions of authors' productivity for some in that discipline (e.g., Sigelman, 2008), though not for others (e.g., Cooper & Jacoby, 2008;Schneider & Jacoby, 2008).

- **Match 3** (sim 0.398) [body/sentence] Academic Ghost Authorship.: In addition to the obscuring of the true authorship of these works, the extent to which the writing encourages bias toward a particular product or point of view emerges as a concern.

Reference [20]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.1907.11692

Resolved title: RoBERTa: A Robustly Optimized BERT Pretraining Approach

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.1907.11692.pdf

Raw reference: [20] Yinhan Liu et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach. 2019. arXiv: 1907.11692 [cs.CL]. url: <https://arxiv.org/abs/1907.11692>.

Citing sentences and best evidence

1. Label: related **sim:** 0.605 **hint:** unclear **zone:** body

Citing: XLM-R[9] extended this idea with a much larger training corpus, CC-100, and adopted RoBERTa[20]-style training improvements such as dynamic masking and larger batches.

- **Match 1** (sim 0.605) [body/sentence] Results: 9 Finally, we pretrain RoBERTa for significantly longer, increasing the number of pretraining steps from 100K to 300K, and then further to 500K.
- **Match 2** (sim 0.597) [body/sentence] Related Work: Our improved pretraining procedure, which we call RoBERTa, achieves state-of-the-art results on GLUE, RACE and SQuAD, without multi-task finetuning for GLUE or additional data for SQuAD.
- **Match 3** (sim 0.581) [body/paragraph] Introduction: We present a replication study of BERT pretraining (Devlin et al., 2019), which includes a careful evaluation of the effects of hyperparameter tuning and training set size. We find that BERT was significantly undertrained and propose an improved recipe for training BERT models, which we call RoBERTa, that can match or exceed the performance of all of the post-BERT methods. Our modifications are simple, they include: (1) training the model longer,...

Reference [21]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/2023.emnlp-main.813

Resolved title: XLM-V: Overcoming the Vocabulary Bottleneck in Multilingual Masked Language Models

Download status: downloaded

Downloaded PDF: pdfs\10.18653_v1_2023.emnlp-main.813.pdf

Raw reference: [21] Davis Liang et al. XLM-V: Overcoming the Vocabulary Bottleneck in Multilingual Masked Language Models. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore, Dec. 2023. doi: 10.18653/v1/2023.emnlp-main. 813. url: <https://aclanthology.org/2023.emnlp-main.813/>.

Citing sentences and best evidence

1. Label: related **sim:** 0.546 **hint:** unclear **zone:** abstract

Citing: Newer families such as XLM-V[21], ModernBERT[22], and mmBERT[23] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.546) [abstract/sentence] abstract: Tokenizations using our vocabulary are typically more semantically meaningful and shorter compared to XLM-R. Leveraging this improved vocabulary, we train XLM-V, a multilingual language model with a one million token vocabulary.

- **Match 2** (sim 0.545) [body/sentence] Building larger vocabularies: Our work seeks to combine and improve upon existing methods for building large-scale vocabularies, pretrain with substantially bigger datasets, and explore vocabularies of 1M tokens and beyond.

- **Match 3** (sim 0.544) [body/sentence] Sentencepiece: This method is used by both XLM-R and our work. et al. (2020) proposed an approach to multilingual vocabulary construction that balances the trade-off between optimizing for cross-lingual subword sharing and the need for robust representation of individual languages.

Reference [22]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2412.13663

Resolved title: Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2412.13663.pdf

Raw reference: [22] Benjamin Warner et al. "Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference". In: (2024). arXiv: 2412.13663 [cs.CL]. url: <https://arxiv.org/abs/2412.13663>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.645 **hint:** body **zone:** body

Citing: Newer families such as XLM-V[21], ModernBERT[22], and mmBERT[23] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.645) [body/paragraph] Introduction: Recent modernization efforts have only partially addressed the shortcomings of encoder-only models due to limited breadth. MosaicBERT (Portes et al., 2023), CrammingBERT (Geiping and Goldstein, 2023), and AcademicBERT (Izsak et al., 2021) focused on matching BERT performance with better training efficiency. NomicBERT (Nussbaum et al., 2024) and GTE-en-MLM (Zhang et al., 2024) (developed concurrently to this work) introduced longer-context encod...

- **Match 2** (sim 0.600) [body/sentence] Conclusion: ModernBERT has a native sequence length of 8,192 tokens and incorporates recent architecture improvements, such as GeGLU layers, RoPE positional embeddings, and alternating local-global attention.

- **Match 3** (sim 0.586) [body/paragraph] Downstream Results and Discussion: Short-Context Retrieval On BEIR, both variants of ModernBERT outperform existing encoders in both the DPR and ColBERT settings, including the recent GTE-en-MLM and NomicBERT models designed to serve as better backbones for retrieval (Zhang et al., 2024; Nussbaum et al., 2024) ModernBERT-large increases its lead despite having comparatively fewer parameters at 395M to GTE-en-MLM-large's 435M.

Reference [23]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2509.06888

Resolved title: mmBERT: A Modern Multilingual Encoder with Annealed Language Learning

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2509.06888.pdf

Raw reference: [23] Marc Marone et al. mmBERT: A Modern Multilingual Encoder with Annealed Language Learning. 2025. arXiv: 2509.06888 [cs.CL]. url: <https://arxiv.org/abs/2509.06888>.

Citing sentences and best evidence

1. Label: supported **sim:** 0.628 **hint:** body **zone:** body

Citing: Newer families such as XLM-V[21], ModernBERT[22], and mmBERT[23] aim to improve multilingual coverage, tokenizer behaviour, or runtime efficiency.

- **Match 1** (sim 0.628) [body/sentence] Conclusion: MMBERT improves over the previous generation of multilingual encoders such as XLM-R while being a drop-in replacement.
- **Match 2** (sim 0.625) [body/sentence] Introduction: MMBERT improves over XLM-R across the board, and even beats models like OpenAI's o3 (OpenAI, 2025) and Google's Gemini 2.5 Pro (Comanici et al., 2025) on low-resource languages.
- **Match 3** (sim 0.624) [body/paragraph] Introduction: MMBERT improves over XLM-R across the board, and even beats models like OpenAI's o3 (OpenAI, 2025) and Google's Gemini 2.5 Pro (Comanici et al., 2025) on low-resource languages. We show that including the low-resource languages in the decay phase enables rapid learning, boosting performance on these languages roughly 2x despite only using 100B tokens. Overall, MMBERT is the first model to show significant improvements over XLM-R for massively mu...

Reference [24]

Occurrences in work: 2

Resolved DOI: 10.1007/s10579-024-09759-3

Resolved title: PESTS: Persian_English cross lingual corpus for semantic textual similarity

Download status: abstract_saved

Raw reference: [24] Mohammad Abdous, Poorya Piroozfar, and Behrouz MinaeiBidgoli. "PESTS: Persian_English cross lingual corpus for semantic textual similarity". In: Language Resources and Evaluation 59.2 (2025), pp. 1179–1199. issn: 15740218. doi: 10.1007/s10579-024-09759-3. url: <https://doi.org/10.1007/s10579-024-09759-3>.

Citing sentences and best evidence

1. Label: related **sim:** 0.551 **hint:** weak_abstract **zone:** abstract

Citing: Recent studies report strong results for transformer-based multilingual embeddings in semantic comparison and paraphrase-oriented cross-lingual settings[24].

- **Match 1** (sim 0.551) [abstract/paragraph] abstract: In recent years, there has been significant research interest in the subtask of natural language processing called semantic textual similarity. Measuring semantic similarity between words or terms, sentences, paragraphs and documents plays an important role in natural language processing and computational linguistics. It finds applications in question-answering systems, semantic search, fraud detection, machine translation, information retrieval...

- **Match 2** (sim 0.515) [abstract/sentence] abstract: In recent years, there has been significant research interest in the subtask of natural language processing called semantic textual similarity.

- **Match 3** (sim 0.485) [abstract/sentence] abstract: Due to the lack of available cross-lingual semantic similarity datasets, many current models in this field rely on machine translation.

2. Label: related **sim:** 0.490 **hint:** weak_abstract **zone:** abstract

Citing: This is suitable for cross-lingual verification because it does not rely on exact wording overlap and does not assume literal translation[24].

- **Match 1** (sim 0.490) [abstract/sentence] abstract: To achieve cross-lingual semantic similarity, it is essential to have corpora that consist of sentence pairs in both the source and target languages.

- **Match 2** (sim 0.486) [abstract/sentence] abstract: However, this dependence on machine translation can result in reduced model accuracy due to the potential propagation of translation errors.

- **Match 3** (sim 0.480) [abstract/sentence] abstract: Due to the lack of available cross-lingual semantic similarity datasets, many current models in this field rely on machine translation.

Reference [25]

Occurrences in work: 1

Resolved DOI: 10.32672/jnkti.v8i1.8790.s1099

Resolved title: Turnitin

Download status: downloaded

Downloaded PDF: pdfs\10.32672_jnkti.v8i1.8790.s1099.pdf

Raw reference: [25] LLC Turnitin. Turnitin. 2025. url: <https://www.turnitin.com/> (visited on 09/21/2025).

Verification status: empty_reference_index

Reference [26]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [26] LLC Turnitin. iThenticate. 2025. url: <https://www.ithenticate.com/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [27]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [27] Fakulta informatiky Masarykovy univerzity – Vývojový tým IS MU. Odevzdej CZ. 2025.
url: <https://odevzdej.cz/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [28]

Occurrences in work: 2

Download status: no_valid_doi

Raw reference: [28] CVTI SR. Centrálny register záverečných a kvalifikačných prác. 2025. url: <https://crzp.cvtisr.sk/> (visited on 09/21/2025).

Verification status: no_source_text

Reference [29]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/n19-1361

Resolved title: Structural Scaffolds for Citation Intent Classification in Scientific Publications

Download status: downloaded

Downloaded PDF: pdfs\10.18653_v1_n19-1361.pdf

Raw reference: [29] Arman Cohan et al. "Structural Scaffolds for Citation Intent Classification in Scientific Publications". In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). Ed. by Jill Burstein, Christy Doran, and Thamar Solorio. Minneapolis, Minnesota: Literature Association for Computational Linguistics, June 2019, pp. 3586–3596. doi: 10.18653/v1/N19-1361. url: <https://aclant...>

Citing sentences and best evidence

1. Label: related **sim:** 0.594 **hint:** unclear **zone:** body

Citing: Some research focuses on citation intent and rhetorical function, as in SciCite[29].

- **Match 1** (sim 0.594) [body/section] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The...

- **Match 2** (sim 0.594) [body/paragraph] Introduction: Citations play a unique role in scientific discourse and are crucial for understanding and analyzing scientific work (Luukkonen, 1992;Leydesdorff, 1998). They are also typically used as the main measure for assessing impact of scientific publications, venues, and researchers (Li and Ho, 2008). The nature of citations can be different. Some citations indicate direct use of a method while some others merely serve as acknowledging a prior work. The...

- **Match 3** (sim 0.562) [body/sentence] Structural scaffolds: In scientific writing there is a connection between the structure of scientific papers and the intent of citations.

Reference [30]

Occurrences in work: 1

Resolved DOI: 10.1287/inte.1070.0317

Resolved title: The Ombudsman: Verification of Citations: Fawltly Towers of Knowledge?

Download status: abstract_saved

Raw reference: [30] Malcolm Wright and J. Scott Armstrong. "The Ombudsman: Verification of Citations: Fawltly Towers of Knowledge?" In: Interfaces 38.2 (Apr. 2008), 125–139. issn: 1526-551X. doi: 10.1287/inte.1070.0317. url: <http://dx.doi.org/10.1287/inte.1070.0317>.

Citing sentences and best evidence

1. Label: related **sim:** 0.569 **hint:** weak_abstract **zone:** abstract

Citing: Wright and Armstrong[30] described citation accuracy as a structural weakness in scholarly communication.

- **Match 1** (sim 0.569) [abstract/sentence] abstract: The prevalence of faulty citations impedes the growth of scientific knowledge.
- **Match 2** (sim 0.522) [abstract/paragraph] abstract: The prevalence of faulty citations impedes the growth of scientific knowledge. Faulty citations include omissions of relevant papers, incorrect references, and quotation errors that misreport findings. We discuss key studies in these areas. We then examine citations to "Estimating nonresponse bias in mail surveys," one of the most frequently cited papers from the Journal of Marketing Research, to illustrate these issues. This paper is especially...
- **Match 3** (sim 0.475) [abstract/sentence] abstract: Faulty citations include omissions of relevant papers, incorrect references, and quotation errors that misreport findings.

Reference [31]

Occurrences in work: 1

Resolved DOI: 10.1197/jamia.m1683

Resolved title: Accuracy of References in Five Biomedical Informatics Journals

Download status: download_failed

Raw reference: [31] Dominik Aronsky, Joel Ransom, and Kevin Robinson. "Accuracy of References in Five Biomedical Informatics Journals". In: Journal of the American Medical Informatics Association 12.2 (Mar. 2005), pp. 225–228. issn: 10675027. doi: 10.1197/jamia.M1683. eprint: <https://academic.oup.com/jamia/article-pdf/12/2/225/2321813/12-2-225.pdf>. url: <https://doi.org/10.1197/jamia.M1683>.

Verification status: no_source_text

Reference [32]

Occurrences in work: 1

Resolved DOI: 10.1080/0361526x.2024.2433256

Resolved title: The Problem with False Positives: AI Detection Unfairly Accuses Scholars of AI Plagiarism

Download status: download_failed

Raw reference: [32] Louie Giray. "The Problem with False Positives: AI Detection Unfairly Accuses Scholars of AI Plagiarism". In: The Serials Librarian 85.5-6 (2024), pp. 181–189. doi: 10.1080/0361526X.2024.2433256. eprint: <https://doi.org/10.1080/0361526X.2024.2433256>. url: <https://doi.org/10.1080/0361526X.2024.2433256>.

Verification status: no_source_text

Reference [33]

Occurrences in work: 2

Resolved DOI: 10.48550/arxiv.2505.20158

Resolved title: Evaluating Software Plagiarism Detection in the Age of AI: Automated Obfuscation and Lessons for Academic Integrity

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2505.20158.pdf

Raw reference: [33] Timur Sa■lam and Larissa Schmid. Evaluating Software Plagiarism Detection in the Age of AI: Automated Obfuscation and Lessons for Academic Integrity. 2025. arXiv: 2505.20158 [cs.SE]. url: <https://arxiv.org/abs/2505.20158>.

Citing sentences and best evidence

1. Label: related **sim:** 0.566 **hint:** unclear **zone:** body

Citing: Recent work also warns that threshold choices and dataset bias can distort reported performance, especially in imbalanced settings[32][33].

- **Match 1** (sim 0.566) [body/sentence] Internal Validity: Public datasets often contain incomplete or biased plagiarism labels.
- **Match 2** (sim 0.546) [body/sentence] Similarity Metrics: Due to varying similarity distributions for different datasets, they can only be set after seeing the results, thus introducing a strong bias.
- **Match 3** (sim 0.532) [body/paragraph] Similarity Metrics: A common anti-pattern in existing works is evaluating plagiarism detectors using a fixed similarity threshold where scores above count as successful detections, and those below do not. While this simplifies deriving precision, recall, and F1 scores, this approach is fundamentally flawed, as the threshold can arbitrarily influence results and it can be tuned to favor one approach. Since no universal threshold fits all datasets, thresholds are ch...

2. Label: related **sim:** 0.588 **hint:** unclear **zone:** body

Citing: Similarity thresholds also have to be calibrated carefully, since poor calibration can make the output look more certain than it really is[33].

- **Match 1** (sim 0.588) [body/paragraph] Similarity Metrics: A common anti-pattern in existing works is evaluating plagiarism detectors using a fixed similarity threshold where scores above count as successful detections, and those below do not. While this simplifies deriving precision, recall, and F1 scores, this approach is fundamentally flawed, as the threshold can arbitrarily influence results and it can be tuned to favor one approach. Since no universal threshold fits all datasets, thresholds are ch...
- **Match 2** (sim 0.561) [body/sentence] Similarity Metrics: Since no universal threshold fits all datasets, thresholds are chosen arbitrarily.
- **Match 3** (sim 0.547) [body/sentence] Similarity Metrics: While this simplifies deriving precision, recall, and F1 scores, this approach is fundamentally flawed, as the threshold can arbitrarily influence results and it can be tuned to favor one approach.

Reference [34]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [34] Crossref. Crossref. 2026. url: <https://www.crossref.org/>.

Verification status: no_source_text

Reference [35]

Occurrences in work: 1

Download status: no_valid_doi

Raw reference: [35] GROBID. <https://github.com/kermitt2/grobid>. 2008–2026. swb: 1: dir:dab86b296e3c3216e2241968f0d63b68e8209d3c.

Verification status: no_source_text

Reference [36]

Occurrences in work: 1

Resolved DOI: 10.1007/978-3-030-96957-8_34

Resolved title: Identifying Machine-Paraphrased Plagiarism

Download status: downloaded

Downloaded PDF: pdfs\10.1007_978-3-030-96957-8_34.pdf

Raw reference: [36] Jan Philip Wahle et al. "Identifying Machine-Paraphrased Plagiarism". In: Information for a Better World: Shaping the Global Future: 17th International Conference, IConference 2022, Virtual Event, February 28 – March 4, 2022, Proceedings, Part I. Springer-Verlag, 2022, 393–413. doi: 10.1007/978-3-030-96957-8_34. url: https://doi.org/10.1007/978-3-03096957-8_34.

Citing sentences and best evidence

1. Label: related **sim:** 0.571 **hint:** unclear **zone:** body

Citing: Passage-level comparison is therefore more informative than whole-document similarity[36].

- **Match 1** (sim 0.571) [body/sentence] Quantitative Evaluation: -based methods tend to have a lower similarity compared to the text-based methods, whereby the highest Jaccard index between two citation-based methods is achieved for Walklets and
- **Match 2** (sim 0.569) [body/sentence] DISCUSSION: The analysis of recommendation similarity also shows little overlap between the citation-based methods and the text-based methods (Figure 3).
- **Match 3** (sim 0.543) [body/sentence] RELATED WORK: Also, Bhattacharya et al. [6] compare citation similarity methods, i.e., Bibliographic Coupling, Co-citation, Dispersion [33] and Node2Vec [17]), and text similarity methods like Paragraph Vectors.

Reference [37]

Occurrences in work: 1

Resolved DOI: 10.48550/arxiv.2006.07264

Resolved title: Low-resource Languages: A Review of Past Work and Future Challenges

Download status: downloaded

Downloaded PDF: pdfs\10.48550_arxiv.2006.07264.pdf

Raw reference: [37] Alexandre Magueresse, Vincent Carles, and Evan Heetderks. Low-resource Languages: A Review of Past Work and Future Challenges. June 2020. doi: 10.48550/arXiv.2006.07264.

Citing sentences and best evidence

1. Label: related **sim:** 0.614 **hint:** unclear **zone:** abstract

Citing: Another weakness is its dependence on pretrained multilingual models, whose performance may be lower for low-resource languages or for highly specialised academic language[37].

- **Match 1** (sim 0.614) [abstract/sentence] abstract: A current problem in NLP is massaging and processing low-resource languages which lack useful training attributes such as supervised data, number of native speakers or experts, etc.

- **Match 2** (sim 0.595) [body/section] Introduction: In the 1990s, the tools of Natural Language Processing (NLP) experienced a major shift, transitioning from rule-based approaches to statisticalbased techniques. Most of today's NLP research focuses on 20 of the 7000 languages of the world, leaving the vast majority of languages understudied. These languages are often referred to as low-resource languages (LRLs), ill-defined as this qualifier can be. LRLs can be understood as less studied, resour...

- **Match 3** (sim 0.591) [body/sentence] Classification: From a machine learning standpoint, one alternative is adopting lighter, less resource-dependent baseline models that are still successful with the English language and modifying them to maximize success with a particular LRL.

Reference [38]

Occurrences in work: 1

Resolved DOI: 10.18653/v1/d19-1371

Resolved title: SciBERT: A Pretrained Language Model for Scientific Text

Download status: download_failed

Raw reference: [38] Iz Beltagy, Kyle Lo, and Arman Cohan. “SciBERT: A Pretrained Language Model for Scientific Text”. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. 2019, pp. 3615–3620. doi: 10.18653/v1/D19-1371.

Verification status: no_source_text

Reference [39]

Occurrences in work: 1

Resolved DOI: 10.47191/ijsshr/v7-i10-39

Resolved title: Challenges of Translating Idiomatic Expressions: A Cross-Linguistic Analysis at a University in Hanoi, Vietnam

Download status: downloaded

Downloaded PDF: pdfs\10.47191_ijsshr_v7-i10-39.pdf

Raw reference: [39] Dang Chung. "Challenges of Translating Idiomatic Expressions: A CrossLinguistic Analysis at a University in Hanoi, Vietnam". In: International Journal of Social Science and Human Research 07 (2024). doi: 10.47191/ ijsshr/v7-i10-39. 14 Scientific Part of the Thesis TBD Appendices A Appendices A Appendices A-1 B Work Plan B Work Plan B-1 C Technical Documentation C Technical Documentation C-1 D Contents of the Digital Medium D Contents of the Digital Medium

Citing sentences and best evidence

1. Label: no_support **sim:** 0.288 **hint:** unclear **zone:** abstract

Citing: Similarity scores may be affected by translation differences, paraphrasing, domain-specific terminology, or idiomatic expressions[39].

- **Match 1** (sim 0.288) [abstract/sentence] abstract: The right to use all or part of this article, including tables and figures in future works of their own, provided that the proper acknowledgment is made to the Publisher as copyright holder, and ■

- **Match 2** (sim 0.273) [abstract/sentence] abstract: In connection with this assignment, the Author acknowledges that IJSSHR will have the right to print, publish, and create derivative works throughout the world, all rights in and to all revisions or versions or subsequent editions of the Work in all languages and media throughout the world.

- **Match 3** (sim 0.263) [abstract/paragraph] abstract: This transfer of copyright gives IJSSHR the right to develop, promote, distribute, and archive a body of scientific works throughout the world. [3] The Author hereby grants and assigns to IJSSHR all rights in and to Author's work in and contributions to the Work. In connection with this assignment, the Author acknowledges that IJSSHR will have the right to print, publish, and create derivative works throughout the world, all rights in and to all...